From Agriculture to Services: Development without Industrialization

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Abstract

This paper studies the long-run consequences of developing without industrialization. Countries with higher historical peaks in manufacturing employment grow faster, accumulate more tertiary education, experience faster fertility declines, and develop more skill-intensive service sectors. The same pattern appears within the United States: counties with larger manufacturing booms later developed service occupations with higher educational content. I interpret these facts through a unified growth model with structural change, endogenous fertility, education, and heterogeneous producer services. In the model, manufacturing creates demand for high-skill organizational tasks that build the skill and knowledge base of the service economy that follows. Calibrated to the United States over 1850–2019, the model explains 31 percent of cross-country log GDP per capita gaps, 60 percent of GDP per capita growth differences, 53 percent of tertiary education differences, and 39 percent of cumulative population growth differences. Countries can move into services without industrializing, but the service economy they build is less skill-intensive and grows more slowly.

Keywords: structural change, service-led growth, industrialization, unified growth, fertility, human capital.

JEL Classification: O14, O15, O41, J13, J23, O33, E24.

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1 Introduction

Modern economic growth began with industrialization. The British Industrial Revolution transformed a largely agrarian economy into one organized around manufacturing, mechanized production, and increasingly complex networks of suppliers, managers, clerks, engineers, accountants, and other producer-service occupations. In the vast majority of today's rich countries, the long-run movement of labor out of agriculture was accompanied by a substantial manufacturing phase before services came to dominate employment. Yet this path has become increasingly uncommon. Many developing countries now move directly from agriculture into services, reaching lower manufacturing employment peaks and deindustrializing at lower income levels than earlier industrializers (Rodrik, 2016).

Can countries develop without manufacturing? A narrow answer is yes: labor can leave agriculture and be absorbed by services. But this paper argues that the path out of agriculture matters for the kind of service economy that emerges. Manufacturing is not only a sector that produces goods. It is also a source of demand for high-skill producer services: accounting, management, technical supervision, engineering, logistics, purchasing, quality control, and other coordination tasks. These tasks help organize production and, once created, can become the foundation for a high-skill service economy. By contrast, economies that bypass manufacturing may enter services earlier, but with employment concentrated in low-skill consumer-facing activities such as retail, restaurants, hospitality, and personal services. This distinction matters because manufacturing does not just delay the rise of services; it changes their composition, and through that channel affects education, fertility, and long-run growth.

I provide evidence for this mechanism in two steps. First, I document cross-country facts linking the size of the manufacturing phase to later development outcomes. Countries with higher peak manufacturing employment shares grow faster after they have largely transitioned into services, accumulate more tertiary education, experience faster fertility declines, and develop more skill-intensive service sectors. These patterns hold among economies with high service employment shares, suggesting that manufacturing matters not only because it absorbs labor during the transition, but because it shapes the service economy that follows. Second, I use full-count U.S. Census data from 1850 to 1940 to study the mechanism within a single economy. Counties with larger manufacturing booms later developed service occupations with higher educational content, measured using occupation-level education scores from the 1950 Census. The effects appear after the manufacturing boom, are strongest for counties with the largest manufacturing peaks, and persist for decades. IV estimates based on geography-driven manufacturing exposure point to the same conclusion.

Building on this evidence, I build a unified growth model with structural change, fertility, education, and heterogeneous producer services. In the model, manufacturing raises demand for organizational and technical services that are intensive in skilled labor. This raises the return to education and builds a stock of codified knowledge that persists after manufacturing declines. Economies with larger manufacturing phases therefore enter the service economy with more human capital, lower fertility, and a more productive producer-service sector. Calibrated to the U.S. transition from 1850 to 2019, the model matches the decline of agriculture, the manufacturing hump, the rise of services, the fertility transition, and the high-skill content of producer-service inputs. Counterfactual economies with smaller manufacturing peaks reach lower income, grow more slowly, accumulate less education, have higher population growth, and develop less skill-intensive services. Quantitatively, the mechanism explains 36 percent of cross-country log GDP per capita gaps, 69 percent of GDP per capita growth differences, 67 percent of tertiary education differences, and 51 percent of cumulative population growth differences.

This paper contributes to three literatures. First, it contributes to work on structural transformation and premature deindustrialization. Classic and modern theories of structural change study why labor moves from agriculture to manufacturing and services, emphasizing non-homothetic preferences, sectoral productivity differences, and relative prices (Kuznets, 1966, 1973; Lewis, 1954; Rostow, 1959; Matsuyama, 1992; Kongsamut et al., 2001; Ngai and Pissarides, 2007; Herrendorf et al., 2014; Boppart, 2014; Comin et al., 2021). Recent work studies why manufacturing peaks differ across countries, emphasizing premature deindustrialization, trade, global specialization, technology gaps, and heterogeneous sectoral productivity growth (Rodrik, 2016; Matsuyama, 2009; Uy et al., 2013; Sposi et al., 2021; Huneeus and Rogerson, 2024; Fujiwara and Matsuyama, 2024). I focus instead on the consequences of these differences. I ask whether missing the manufacturing phase changes the development path itself by altering skill demand, fertility choices, knowledge accumulation, and the long-run composition of services.

Second, the paper contributes to work on service-led growth and the organization of production. A growing literature asks whether services can substitute for manufacturing as an engine of development and emphasizes that services are highly heterogeneous (Fan et al., 2023; Peters et al., 2026; Herrendorf et al., 2022; Buera and Kaboski, 2012a,b; Buera et al., 2022; Duernecker and Herrendorf, 2022; Duernecker et al., 2024). Other work shows that development involves the reorganization of production toward larger, more skill-intensive firms and white-collar tasks (Engbom et al., 2025). My contribution is to connect this organizational margin to the sectoral path of development. Manufacturing matters not simply because it has high productivity growth, but because it helps determine whether the service economy expands into high-skill producer and professional services or into low-skill consumer-facing services. This mechanism also connects the paper to work on urbanization without industrialization and consumption cities, where labor leaves agriculture without necessarily entering modern manufacturing (Gollin et al., 2016; Jedwab et al., 2022).

Third, the paper contributes to unified growth theory. Canonical unified growth models explain how technological progress, human-capital accumulation, and fertility decline interact in the transition from stagnation to modern growth (Galor and Weil, 2000; Hansen and Prescott, 2002; Galor, 2005, 2011; Becker et al., 1990; Doepke, 2004; Galor and Moav, 2004). I embed this transition in a model of structural change across agriculture, manufacturing, and services. This matters because sectors differ in their demand for skills and in their contribution to organizational knowledge. The path out of agriculture therefore affects not only sectoral employment shares, but also fertility, education, and the speed at which economies transition to modern growth. In this sense, the paper links two literatures that have often been studied separately: structural change across sectors and the joint evolution of fertility, human capital, and income.

The rest of the paper is structured as follows. Section 2 documents the cross-country facts relating peak manufacturing employment to GDP per capita growth, education, service-sector composition, and fertility. Section 3 uses full-count U.S. Census data to study the effect of historical manufacturing booms on the skill content of services across counties. Section 4 develops the model. Section 5 describes the calibration to the United States over 1850-2019. Section 6 presents the counterfactual results. Section 7 concludes. Appendix A describes the data construction, Appendix B reports additional cross-country evidence, Appendix C reports additional U.S. county evidence, Appendix D contains model details, Appendix E reports calibration details and robustness, and Appendix F describes policy extensions.

2 Cross-Country Evidence

This section documents three cross-country facts linking the size of the manufacturing phase to later development outcomes. Countries with larger manufacturing employment peaks grow faster after moving into services, accumulate more tertiary education, develop more skill-intensive service sectors, and experience faster fertility declines. The analysis combines data on GDP per capita, fertility, population, and broad sectoral employment shares from Gapminder with longer sectoral series from the GGDC 10-Sector Database, tertiary education measures from Barro and Lee (2013), service-sector microdata from IPUMS International, and input-output linkages from the OECD Inter-Country Input-Output tables. The baseline sample includes 174 countries with employment shares over 1990–2018 and long-run GDP per capita, population, and fertility data. Appendix A describes the data sources, sample construction, and variable definitions in detail.

***FACT 1**: Countries with higher peak manufacturing employment shares grow faster as they transition into services.*

The rise of services is often viewed as a drag on aggregate growth. In the standard Baumol cost-disease logic, services experience slower productivity growth than agriculture or manufacturing and as labor reallocates toward them, aggregate productivity growth declines (Baumol, 1967; Duernecker and Sanchez-Martinez, 2023; Duernecker et al., 2024). However, little attention has been given to whether this pattern is different depending on how service economies reach this state, namely whether they went through manufacturing or not.

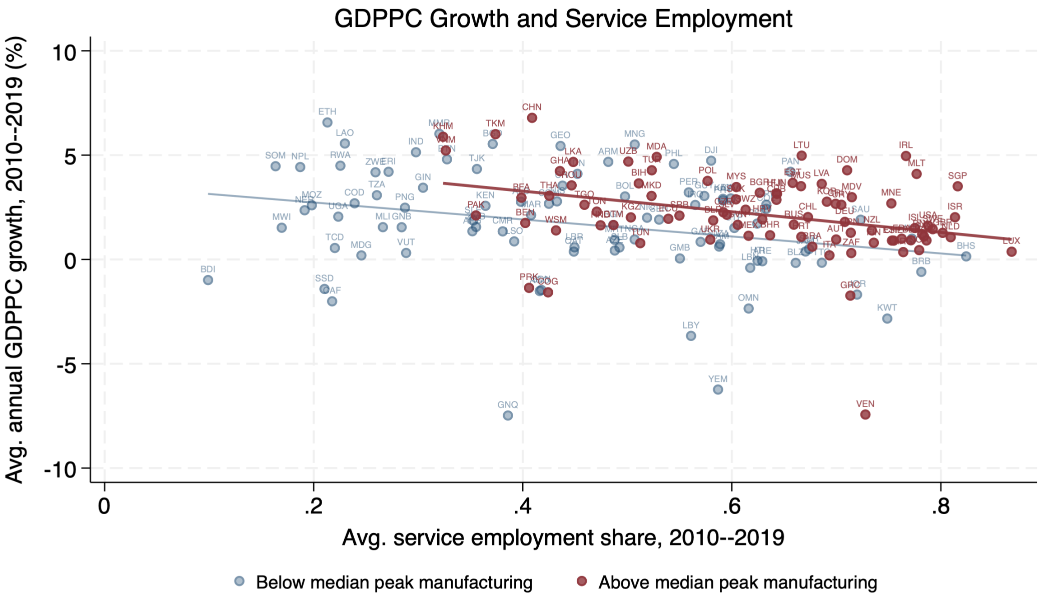


Figure 1: Baumol's cost disease. Countries with above median peak manufacturing employment shares continue to have higher growth rates as they move into services. Data source: ILO and World Bank via Gapminder.

Figure 1 plots the average yearly growth rate of PPP real GDP per capita over 2010-2019 across 174 countries against their average service sector employment share in the same period, distinguishing between countries with above and below median peak manufacturing employment share, which is the maximum employment share in manufacturing over the sample period.[2] While across countries a higher services employment share is associated with a lower aggregate growth rate, those that reached a peak manufacturing employment share above the median of the cross country distribution grow faster at all levels of the services employment share. To get a sense of the magnitude of this relationship I estimate the following simple regression model:

$$y_c = \alpha_r + \alpha_i + \Gamma' X_c + \beta^s \text{Serv. Emp. Share}_c + \beta^m \text{Manu. Peak}_c + \epsilon_c$$

Where y_c is the average growth rate of GDP per capita over 2010-2019, α_i and α_r are, respectively, income quintile and regional fixed effects using the five World Bank region definitions, X_c is a vector of control variables including the institutional quality index from the World Bank's ease of doing business dataset, average GDP per capita over 2010-2019 and years since the peak of manufacturing employment. The results are reported in Table 1. The coefficient on peak manufacturing employment is positive

2. I exclude 2020 due to the effect of the COVID-19 pandemic on output. The peak manufacturing employment share is the maximum observed employment share in manufacturing over 1990-2018 for the countries in the ILOEST data and 1960-2018 for those also appearing in the GGDC 10-Sector database.

Table 1: Dep. Var. is Avg GDPPC growth rate 2010-2019

	(1)	(2)	(3)	(4)
Avg. Serv. Emp. Share, 2010–2019	-2.742***	-5.171***	-7.566***	-9.962***
	(-2.84)	(-4.49)	(-3.59)	(-4.17)
Peak Emp. % Man.		9.658***	5.455**	5.511**
		(4.59)	(2.16)	(2.09)
N	174	174	174	174
R2	0.0469	0.130	0.215	0.284
Income Quintile FE	no	no	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

and statistically significant across specifications, and its magnitude is economically meaningful. In the most saturated specification, a one percentage point higher manufacturing peak is associated with an offset of more than half of the growth slowdown associated with a one percentage point higher service employment share. As shown below, this relationship is consistent with countries with larger manufacturing peaks accumulating more human capital and developing more skill-intensive services.

FACT 2*: Countries with higher peak manufacturing employment shares accumulate more tertiary education and develop more skill-intensive services.*

The growth patterns in Fact 1 are accompanied by differences in human-capital accumulation. If manufacturing-intensive development is associated with greater demand for skill-intensive tasks, then countries with larger manufacturing phases should also display larger increases in education and a more skill-intensive service sector. Figure 2 plots tertiary education against GDP per capita, separately for countries above and below the median peak manufacturing employment share. Tertiary education rises with income in both groups, but the relationship is steeper among countries with larger manufacturing peaks. This cross-sectional pattern also appears within countries over time. To summarize the relationship, I estimate

$$\Delta e_c = \alpha + \gamma \Delta \log y_c + \sum_{q=2}^{5} \beta_q \, \mathbb{1}\left\{\text{Manu. Peak}_c \in Q_q\right\} + \varepsilon_c.$$

Here $\Delta e_c = e_{c,T_c} - e_{c,t_c}$ is the change in tertiary education for country c between the first and last years in which both education and GDP per capita are observed over 1960–2020, and $\Delta \log y_c = \log y_{c,T_c} - \log y_{c,t_c}$ is the corresponding change in log GDP per capita. The omitted category is the lowest peak-manufacturing quintile, so the coefficients β_q compare education gains in higher peak-manufacturing quintiles to those in the lowest quintile, conditional on income growth.

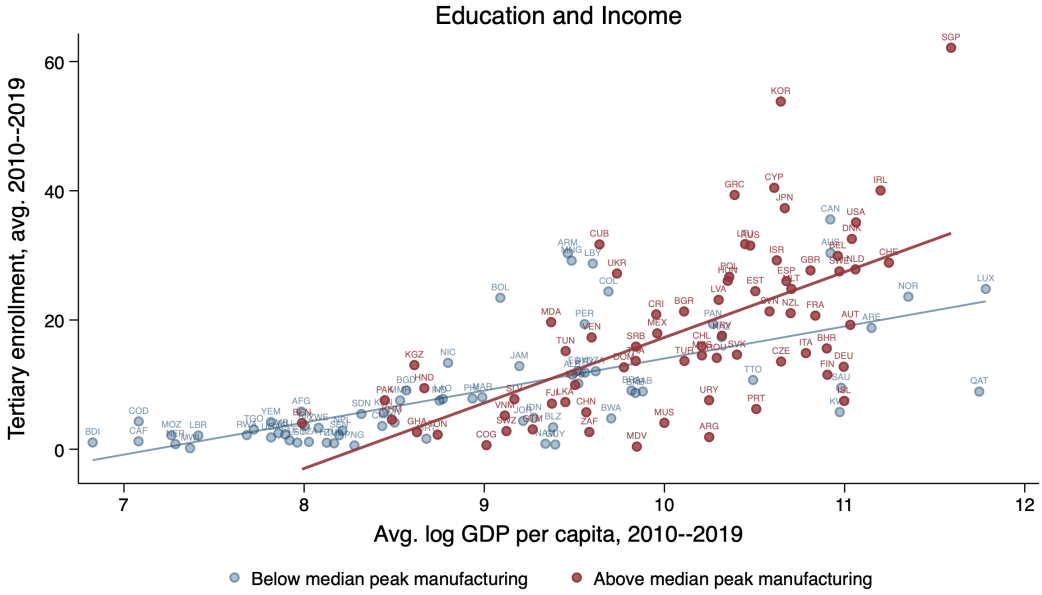


Figure 2: Tertiary education and GDP per capita. Tertiary education rises with development, but more steeply among countries with above-median peak manufacturing employment. Data source: Barro and Lee (2015) and ILO via Gapminder.

Table 2 reports the results. Conditional on income growth, countries with larger manufacturing peaks experience larger increases in tertiary education. In the highest peak-manufacturing quintile, the increase is up to 12.3 percentage points larger than in the lowest quintile. I then ask whether this education pattern is mirrored in the composition of services. Using the 2015 OECD Inter-Country Input-Output table, I compute the service-sector output induced by a \$100 final-demand increase in agriculture, manufacturing, or consumer services:

$$x_j = (I - A)^{-1} f_j,$$

where A is the matrix of technical input coefficients and f_j is the sector-specific final-demand shock. I classify finance and insurance, computer and information services, and professional, scientific, and technical activities as high-skill services, and the remaining service industries as low-skill services. For the definition of consumer services see Appendix B.

Figure 3 reports the resulting service-sector mix. The service output induced by manufacturing contains a larger high-skill component than the service output induced by consumer-service demand. This pattern is consistent with the view that manufacturing is linked to a more skill-intensive part of the service sector, not only through manufacturing employment itself, but also through its intermediate-input linkages. Agriculture also induces some high-skill service activity, but agricultural expenditure shares decline with development (Herrendorf et al., 2013). The input-output evidence therefore suggests that, among the sectors expanding during the transition out of agriculture, manufacturing is more closely connected to the high-skill service activities

Table 2: Education Gains by Peak Manufacturing Quintile

	Change in education
Δ log GDP per capita	5.391**
	(2.215)
Peak manufacturing Q2	5.337**
	(2.160)
Peak manufacturing Q3	4.184**
	(1.919)
Peak manufacturing Q4	7.647***
	(1.979)
Peak manufacturing Q5	12.300***
	(2.315)
Constant	-0.653
	(1.049)
Observations	139
R-squared	0.296

Notes: Robust standard errors in parentheses. The omitted category is Q1, the lowest peak-manufacturing quintile. The dependent variable is the change in education between the first and last usable country-year. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

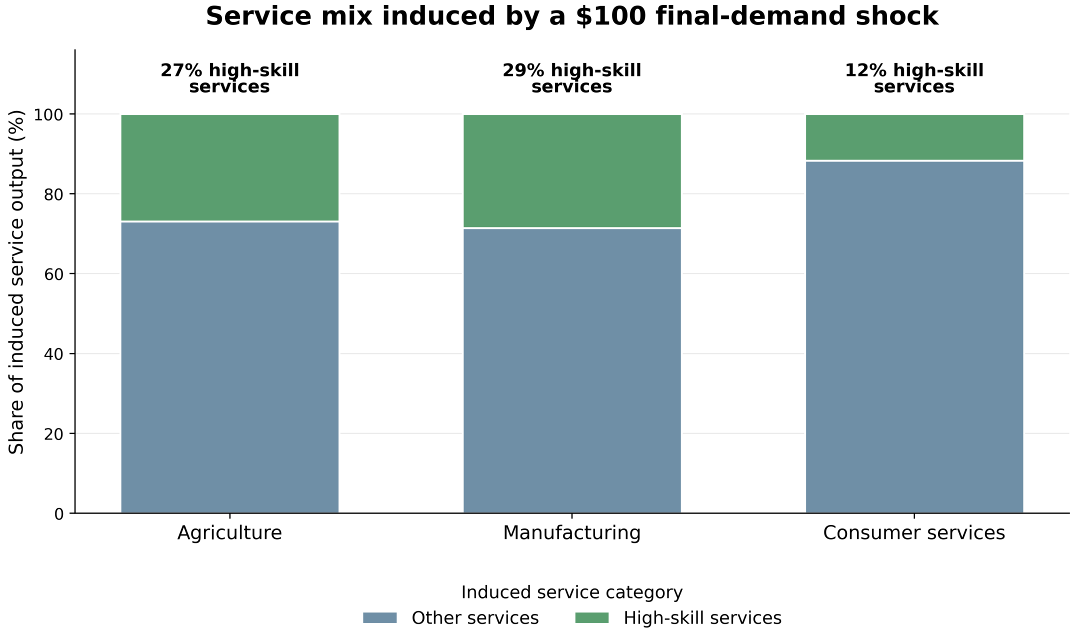


Figure 3: Service-sector output induced by a $100 final-demand increase in agriculture, manufacturing, or consumer services, computed using Leontief total requirements. Consumer services are industries where final consumption exceeds 50 percent of gross output. Data source: 2015 OECD ICIO.

measured here. Appendix B shows that the pattern is similar across country income groups and under broader definitions of consumer-service demand. The education and input-output patterns are consistent with higher skill demand in economies with larger manufacturing phases. I next examine the demographic counterpart of this pattern. In standard quantity-quality models, higher returns to education induce households to invest more in child quality and reduce fertility (Becker et al., 1990; Galor and Weil, 2000; Galor, 2005, 2011). If larger manufacturing phases are associated with higher returns to education, they should therefore also be associated with faster fertility declines.

FACT 3*: Countries with higher peak manufacturing employment shares experience faster demographic transitions, exhibiting lower fertility at comparable levels of development.*

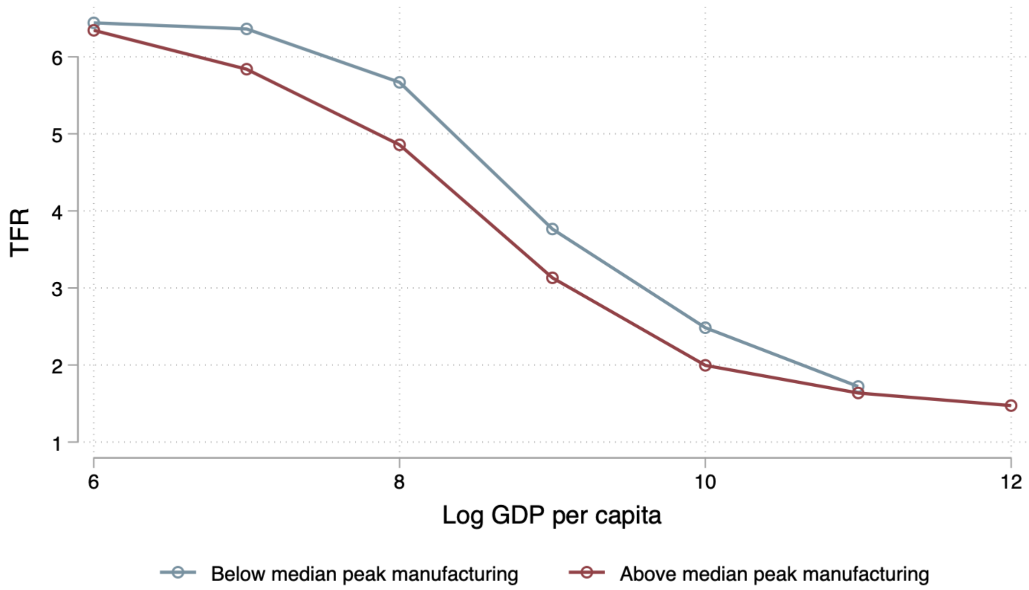


Figure 4: Scatterplot of the Total Fertility Rate (TFR) over log of GDP per capita across 176 countries observed from 1800 to 2020. Countries are grouped by being above and below median peak manufacturing employment share. Data Source: Gapminder.

Figure 4 plots the mean fertility rate over the log of GDP per capita in countries in the above median and below median peak manufacturing employment share groups, using data on fertility and GDP per capita using a balanced cross-country panel back to 1800. Countries with higher peaks had lower fertility rates even at relatively low levels of GDP per capita, for comparison purposes a log GDP per capita of 6.5 corresponds to that of a median low income country today. This implies that even early in the development process, when today's industrialized countries had relatively low income levels, they still had lower fertility levels than a non industrializing country did at a similar level of development. To formalize this analysis, I run the following descriptive regression:

$$\overline{TFR}_c = \alpha + \beta \,\text{Manu. Peak}_c + \gamma_1 TFR_{c,1800} + \gamma_2 \log GDPpc_{c,1800} + \gamma_3 \text{Peak Year}_c + \delta_r + \varepsilon_c.$$

The dependent variable, $\overline{TFR}_c$, is the average fertility rate observed for country c over 1800-2020 and Manu. Peak$_c$ is the country's peak manufacturing employment share. I control for initial fertility, initial log GDP per capita, and the year the country reached its manufacturing peak and regional fixed effects δ_r.

Table 3: Dep. Var. is Average Fertility Rate 1800-2020

	(1)	(2)	(3)	(4)
Peak Emp. % Man.	-1.788*	-1.036***	-1.184***	-0.700***
	(-1.80)	(-2.75)	(-3.81)	(-2.82)
Fertility rate 1800		0.598***	0.387***	0.316***
		(13.97)	(8.70)	(5.80)
log GDP per capita 1800			-0.810***	-0.574***
			(-5.54)	(-4.28)
Year of manufacturing peak			0.00701*	0.00451
			(1.75)	(1.28)
N	189	189	176	176
R2	0.0465	0.513	0.642	0.760
Region FE	no	no	no	yes

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The results are reported in Table 3. Across specifications, the coefficient on peak manufacturing is negative and statistically significant. This relationship is not simply a comparison between rich and poor countries: it remains after controlling for initial income and initial fertility, and after adding region fixed effects. Together with Figure 4, the evidence suggests that countries that ultimately reached higher manufacturing employment peaks tended to have lower fertility at comparable stages of income development. Taken together, the three facts point to a common pattern: economies with larger manufacturing phases later grow faster as service economies, accumulate more human capital, develop more skill-intensive services, and experience faster fertility declines. These cross-country relationships are descriptive, but they motivate the mechanism studied in the rest of the paper. The next section turns to U.S. county evidence, where manufacturing booms can be measured at a more disaggregated level and followed over time.

3 U.S. County Evidence

The cross-country evidence in Section 2 suggests that economies with larger manufacturing phases later develop more skill-intensive service sectors, accumulate more human capital, and experience faster demographic transitions. This section studies the same relationship within a single country, using U.S. counties during the main period of American industrialization. The within-country setting is useful for two reasons. First, it holds fixed many national institutions, aggregate shocks, and measurement differences that complicate cross-country comparisons. Second, local manufacturing booms can be measured over time, allowing me to ask whether counties with larger manufacturing peaks subsequently developed more education-intensive service sectors.

I measure county-level manufacturing exposure using county tabulations from the Census of Manufactures over 1880–1940 compiled by Haines (2001). Since consistent county employment denominators are not available before 1950, I define a county's manufacturing peak as the maximum manufacturing employment-to-population ratio observed over this period. Earlier manufacturing censuses are omitted because of well-known data reliability issues (Vickers and Ziebarth, 2024). I combine these manufacturing data with the full-count U.S. Censuses from 1850 to 1940, restricting the sample to fathers of respondents aged 6 to 20 with non-missing industry and employment information.[3] Appendix C.2 reports complementary panel data evidence showing that, within counties over 1850–1940, higher lagged manufacturing employment shares predict larger fertility declines, larger increases in school enrollment, and larger increases in the education score of service occupations.

The main outcome in this section is the skill content of service employment. Because individual education is not observed before 1950, I use an occupation-based education score. For each three-digit occupation, IPUMS assigns the share of workers in that occupation in 1950 who had completed some higher education. I then average this score across workers employed in service industries in each county-year. The resulting variable, $EDSCOR50_{ct}$, is higher when a county's service sector is composed of occupations that were education-intensive in 1950. This measure captures changes in the occupational composition of services, rather than changes in observed schooling after 1950.

I estimate an event-study specification around each county's manufacturing peak:

$$EDSCOR50_{ct} = \delta_c + \gamma_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - T_c^{peak} = k\} \times \mathbf{1}\{\text{Manu. Peak}_c \in \mathcal{P}\} + \varepsilon_{ct}, \quad (1)$$

where δ_c are county fixed effects, γ_t are census-decade fixed effects, and T_c^{peak} is the decade in which county c reaches its manufacturing peak. The set $\mathcal{P}$ defines groups of counties by the size of the manufacturing peak. The coefficients β_k trace whether

3. I use a common sample to compute county-by-year enrollment rates, education scores, and fertility rates. Appendix A describes the data construction in detail.

counties in a given part of the peak-manufacturing distribution experience differential changes in the skill content of service occupations before and after the manufacturing peak, relative to counties outside that group. I estimate the event study using the imputation estimator of Borusyak et al. (2024) to account for staggered treatment timing.

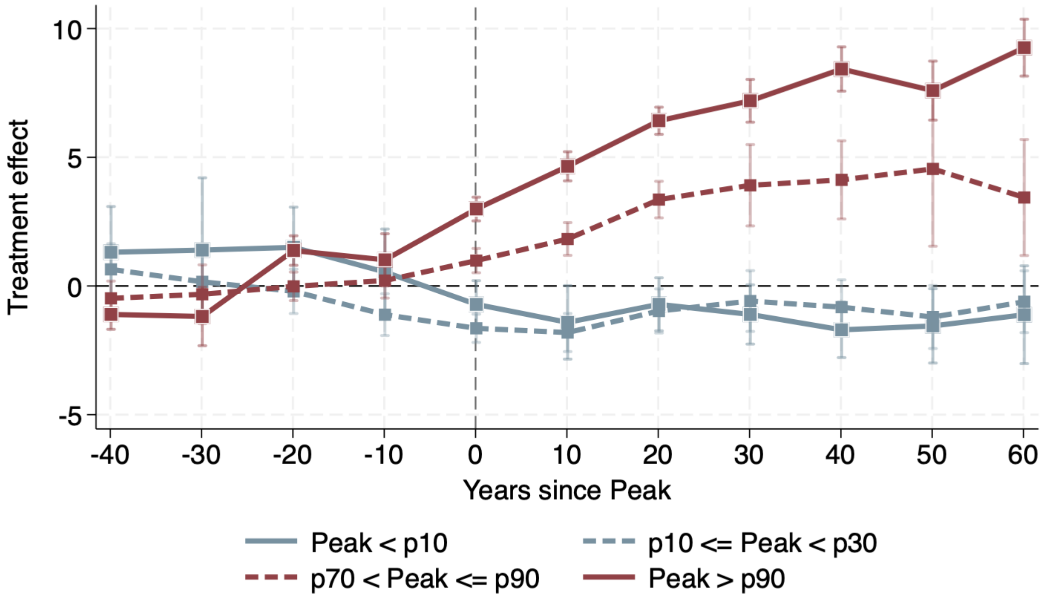


Figure 5: Specification: $Y_{ct} = \delta_c + \gamma_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{\%\text{Peak Man. Emp.} \in \mathcal{P}, t - T_{\text{peak}} = k\} + \epsilon_{ct}$. Y_{ct} is the mean education score of service occupations in county c and decade t; occupation scores equal the 1950 share with ≥ 1 year of college. Sources: U.S. Census (1850–1940); Haines (2010).

Figure 5 reports the event-study estimates. Counties in the top decile of the manufacturing-peak distribution experience a clear increase in the education score of service occupations after the peak. The pre-peak coefficients are close to zero, while the post-peak coefficients rise and remain positive several decades after the peak. The event-study pattern indicates that larger manufacturing peaks lead to a fanning out in the skill intensity of the service sector after the peak, with larger peaks being associated with the rise of a more skilled service sector. Appendix C.3 shows that this conclusion is robust when using finer bins of the manufacturing-peak distribution, and Appendix C.4 decomposes the result by occupation showing that this is primarily driven by professional and technical service sector occupations which are plausibly complementary to the manufacturing sector.

A lingering concern is that counties with large manufacturing peaks may have differed along other dimensions that also shaped later service employment, such as urbanization, market access, or initial human capital. To address this concern, I construct a geography-based predicted manufacturing peak. The prediction uses the fact that the geographic determinants of manufacturing changed over time: proximity to the fall line was especially important for early industrialization because it provided access to water power, while proximity to coal became more important later. I estimate

$$\widehat{\text{Manu. Peak}}_c = \sum_d \Big(\widehat{\beta}_d^C \ln(\text{coal}_c) + \widehat{\beta}_d^F \ln(\text{fall}_c) + \widehat{\beta}_d^O \ln(\text{ohio}_c) + \widehat{\beta}_d^{GL} \text{GreatLakes}_c \Big) \cdot \mathbf{1}\{\text{peak decade} = d\}. \tag{2}$$

The geographic variables are distance to the nearest coal mine active in the late nineteenth century, distance to the Ohio River, distance to the fall line, and an indicator for adjacency to the Great Lakes.[4] I then repeat the event-study exercise by grouping counties according to the predicted component of their manufacturing peak.

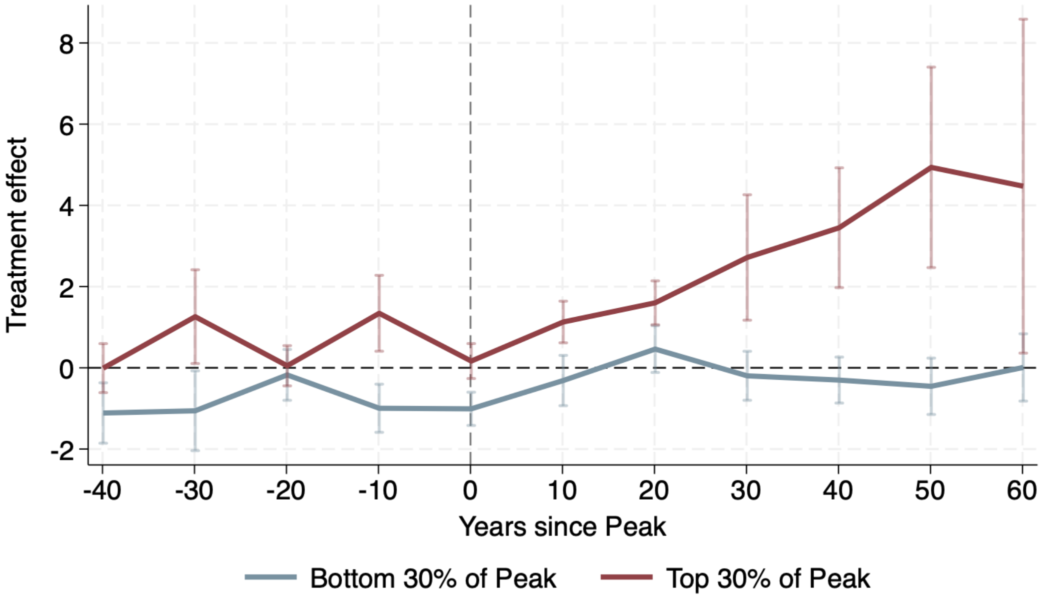


Figure 6: Specification: $Y_{ct} = \delta_c + \gamma_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{\text{Peak \% } \hat{\text{Man}}\text{. Emp.} \in \mathcal{P}, t - T_{\text{peak}} = k\} + \epsilon_{ct}$. Y_{ct} is the mean education score of service occupations in county c and decade t; occupation scores equal the 1950 share with ≥ 1 year of college. Sources: U.S. Census (1850–1940); Haines (2010).

Figure 6 shows that the geography-predicted manufacturing peak produces a similar pattern. Counties predicted to have larger manufacturing peaks experience larger post-peak increases in the education score of service occupations than counties predicted to have smaller peaks. This exercise does not rule out all sources of endogeneity, but it reduces the concern that the baseline pattern is driven by other sources of county variation that are correlated with realized manufacturing peaks.

I next ask whether the relationship between historical manufacturing peaks and skill-intensive local development persists beyond the 1850–1940 census window. Using county outcomes measured in 2000, I estimate

$$y_{c,s,2000} = \delta_s + \beta \, \text{Manu. Peak}_{c,s} + \Gamma' X_{c,s} + \epsilon_{c,s}. \tag{3}$$

4. Data sources and robustness exercises excluding the Ohio River and Great Lakes variables are described in Appendix A and Appendix C.

Here $y_{c,s,2000}$ is a county outcome measured in 2000 and δ_s is a state fixed effect. The variable Manu. $\text{Peak}_{c,s}$ is the county's historical manufacturing peak over 1850–1950. The vector $X_{c,s}$ includes baseline historical controls measured in 1850: log population, urbanization, school enrollment, and the number of colleges.

Table 4: Effect of Peak Manufacturing Employment Share on Outcomes in 2000

	Log patents pc	Frac. college+	Log income	Innovation / total emp.	HS services / total services	Advanced mfg / total mfg
Panel A: OLS						
Peak Emp. % Man.	0.031***	0.127***	0.488***	0.055***	0.156***	0.080*
	(0.006)	(0.036)	(0.105)	(0.021)	(0.052)	(0.042)
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Historical Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	1600	1600	1600	997	997	997
R^2	0.192	0.317	0.401	0.226	0.133	0.300
Panel B: IV						
Peak Emp. % Man.	0.042**	0.421***	0.774	0.190*	0.533*	0.038
	(0.018)	(0.156)	(0.471)	(0.099)	(0.283)	(0.125)
State FE	Yes	Yes	Yes	Yes	Yes	Yes
Historical Controls	Yes	Yes	Yes	Yes	Yes	Yes
Kleibergen–Paap	83.91	83.91	83.91	83.91	83.91	83.91
Observations	1600	1600	1600	997	997	997
R^2	0.189	0.285	0.396	0.181	0.096	0.299

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 reports the results. Counties with larger historical manufacturing peaks have higher patenting, higher college attainment, higher income, larger innovation employment shares, and more high-skill service employment in 2000. Panel B instruments the realized manufacturing peak using the geography-based predicted peak in equation (2). The IV estimates are generally larger than the OLS estimates, though less precise for some outcomes. Appendix A describes the construction of the 2000 outcomes, including patents per capita, innovation employment, high-skill services, and advanced manufacturing. Appendix C.6 shows that counties with larger manufacturing peaks also account for a larger share of university openings over 1800-2000, and Appendix C.7 shows that new manufacturing establishments are followed by increased help-wanted advertisements for skilled manufacturing and skilled service occupations, including technical, administrative, and managerial roles.

Overall, the U.S. evidence mirrors the cross-country patterns: larger manufacturing booms are followed by more education-intensive services and by persistent differences in 2000 outcomes. The next section interprets this evidence through the lens of a growth model linking manufacturing to skill demand, education, fertility, and service composition.

4 Model

The evidence above points to three linked outcomes: economies with larger manufacturing phases later develop more skill-intensive services, accumulate more human capital, and experience faster fertility declines. I now build a model to organize these facts and quantify their long-run implications. The model combines structural change with a unified-growth mechanism. Households choose fertility and education; production takes place in agriculture, manufacturing, and consumer services; and all final-good sectors use producer services as intermediate inputs. Producer services can be performed in low-skill or high-skill mode, and sectors differ in how easily their producer-service tasks can be reorganized into high-skill form.

The central mechanism is that manufacturing creates demand for producer-service tasks that are relatively easy to codify. A larger manufacturing phase therefore raises the demand for skilled labor, expands the stock of codified organizational knowledge, and leaves the economy with a more skill-intensive service sector after manufacturing declines. This affects household choices because higher growth and skill demand raise the return to education, leading parents to invest more in child quality and have fewer children. The model therefore links the size of the temporary manufacturing phase to long-run service composition, human capital, fertility, and growth.

The rest of this section describes the environment, household choices, production, and dynamics. Section 5 takes the model to the U.S. transition, and Section 6 uses the calibrated economy for counterfactuals.

4.1 Environment

Time is discrete. The economy is populated by overlapping generations who live for two periods: childhood and adulthood. During childhood, individuals receive education from their parents and decide whether to become skilled. During adulthood, they work, consume, choose fertility, and invest in the education of their children.

There are three final-good sectors: agriculture, manufacturing, and consumer services. Final goods are consumed by households and produced using sectoral value added, agricultural and manufacturing intermediates, and producer services. Producer services are intermediate inputs used by all final-good sectors. They can be performed in low-skill mode or high-skill mode. High-skill producer services capture professional, managerial, technical, accounting, engineering, planning, and coordination tasks.

The key friction is that sectors differ in how easily their producer-service tasks can be reorganized into high-skill form. In each final sector, tasks are indexed by a codification hurdle. A task is performed in high-skill mode when doing so is cheaper than performing it in low-skill mode. The set of high-skill tasks is therefore endogenous and depends on the relative price of skilled labor and on the stock of codified organizational knowledge H_t. A higher H_t lowers the cost of reorganizing tasks into high-skill form.

Households are indexed by parental origin $i \in \{a, m, s\}$,[5] corresponding to whether the adult generation works in agriculture, manufacturing, or services. Parents allocate time between market work, child rearing, and education of children. Fertility determines population growth, education affects children's human capital, and the skill premium shapes the future supply of skilled labor.

At the end of each period, population, the skilled share, sectoral productivity, and codified organizational knowledge are updated. Children then become adults and make the same choices.

4.2 Households

The household block embeds the quantity-quality mechanism of unified growth theory in a multisector environment. As in Galor and Weil (2000), growth raises the return to human capital by making inherited knowledge less useful. I use the tractable human-capital formulation in Lagerlöf (2006) to discipline this channel quantitatively.

Adults are indexed by sectoral origin $i \in \{a, m, s\}$. A household of origin i chooses final consumption, fertility $n_{i,t}$, and education investment per child $e_{i,t}$. Preferences are

$$U_{i,t} = (1-\gamma)\log C_{i,t} + \gamma \log\left(n_{i,t} h_{i,t}\right), \tag{4}$$

where $C_{i,t}$ is composite consumption and $h_{i,t}$ is the human capital of each child.

Final consumption is aggregated using the non-homothetic CES demand system of Comin et al. (2021):

$$\sum_{j \in \{a,m,c\}} \omega_j^{\frac{1}{\sigma}} \left(\frac{C_{ij,t}}{C_{i,t}^{\varepsilon_j}}\right)^{\frac{\sigma-1}{\sigma}} = 1. \tag{5}$$

Here $C_{ij,t}$ is consumption of final good j, ω_j are sectoral weights, σ is the elasticity of substitution across final goods, and ε_j governs sector-specific income effects. This demand system allows income growth and relative-price movements to jointly determine expenditure reallocation across agriculture, manufacturing, and consumer services.

Manufacturing is the numeraire. The household budget constraint is

$$p_{a,t} C_{ia,t} + C_{im,t} + p_{c,t} C_{ic,t} \le \left[1 - n_{i,t}\left(\tau_i + \tau_e(S_t) e_{i,t}\right)\right] w_t(S_t). \tag{6}$$

The household has one unit of time. Raising a child requires time τ_i, which is allowed to vary by sectoral origin in order to match sectoral fertility differences in the data.[6] Education requires an additional time cost $\tau_e(S_t) e_{i,t}$ per child.

5. I use $i \in \{a, m, s\}$ to index parental origin, where s denotes service-sector origin, and $j \in \{a, m, c\}$ to index final-good sectors, where c denotes consumer services.

6. See Appendix A.

Child human capital is

$$h_{i,t} = \frac{e_{i,t} + \tau_h}{e_{i,t} + \tau_h + g_t}, \tag{7}$$

where g_t is the aggregate growth rate of the economy. This captures the Galor-Weil obsolescence channel in reduced form: when growth is high, inherited knowledge depreciates more quickly, raising the value of education.

The Cobb-Douglas structure of preferences implies simple fertility and education choices. Let $\chi_t \equiv \tau_e(S_t)$. Fertility is

$$n_{i,t} = \frac{\gamma}{\tau_i + \chi_t e_{i,t}}. \tag{8}$$

The education choice is

$$e^*_{i,t} = \max\left\{\sqrt{\frac{g_t(\tau_i - \chi_t \tau_h)}{\chi_t}} - \tau_h,\ 0\right\}. \tag{9}$$

Appendix D derives these expressions. Equations (8) and (9) summarize the demographic channel. Higher growth raises the return to education, increasing $e_{i,t}$. Higher education investment raises the time cost of children and lowers fertility.

Children choose whether to become skilled before entering adulthood. Let

$$\pi_t = \frac{w_{s,t}}{w_{u,t}} \tag{10}$$

denote the skill premium. The probability that a child becomes skilled is

$$p_{s,t} = \frac{1}{1 + \exp\left(-\frac{\beta_\pi \log \pi_t - c_0}{\sigma_{\text{skill}}}\right)}. \tag{11}$$

The parameter β_π governs the responsiveness of skill acquisition to the skill premium, c_0 is the average cost of becoming skilled, and σ_{skill} controls the dispersion of idiosyncratic costs.

4.3 Production and producer services

The production side combines a multisector model of structural transformation with an input-output structure. Following Herrendorf et al. (2013), households consume final expenditure categories, while production is organized through value added and intermediate inputs. This distinction allows services to play two roles: consumer services are final goods, while producer services are intermediate inputs used in all final-good sectors.

There are three final-good sectors $j \in \{a, m, c\}$: agriculture, manufacturing, and consumer services. Value added in sector j is produced using skilled and unskilled labor:

$$VA_{j,t} = A_{j,t} \left[\alpha_j (h_s L_{s,j,t})^{\frac{\sigma_L - 1}{\sigma_L}} + (1 - \alpha_j)(h_u L_{u,j,t})^{\frac{\sigma_L - 1}{\sigma_L}} \right]^{\frac{\sigma_L}{\sigma_L - 1}}. \tag{12}$$

Here $A_{j,t}$ is sectoral value-added productivity, $L_{s,j,t}$ and $L_{u,j,t}$ are skilled and unskilled labor, h_s and h_u are skill-specific efficiency units, α_j governs skill intensity, and σ_L is the elasticity of substitution across skill types.

Gross output is

$$Y_{j,t} = A_{go,j,t} VA_{j,t}^{\theta_{j,t}} M_{a,j,t}^{\gamma_{a,j,t}} M_{m,j,t}^{\gamma_{m,j,t}} PS_{j,t}^{\gamma_{ps,j,t}}, \tag{13}$$

$$\theta_{j,t} + \gamma_{a,j,t} + \gamma_{m,j,t} + \gamma_{ps,j,t} = 1. \tag{14}$$

Here $M_{a,j,t}$ and $M_{m,j,t}$ are agricultural and manufacturing intermediates, and $PS_{j,t}$ is the producer-service composite used by sector j. The input-output shares are allowed to vary over the transition. In the calibration, production begins in 1850 as a value-added technology, $\theta_{j,1850} = 1$, and input-output coefficients then transition toward their observed modern values. This is motivated by evidence that changes in U.S. industry output reflect changes in input-output coefficients as well as final demand (Feldman et al., 1987).

Producer services can be performed in low-skill or high-skill mode:

$$LS_t = A_{ls,t} h_u L_{u,ls,t}, \qquad HS_t = A_{hs,t} h_s L_{s,hs,t}. \tag{15}$$

Low-skill producer services capture routine administrative and support tasks. High-skill producer services capture professional, managerial, technical, accounting, engineering, planning, and coordination tasks.

The key object is the share of producer-service tasks performed in high-skill mode. In each final sector j, producer-service tasks have codification hurdles $z \geq 0$, distributed according to

$$F_j(z) = 1 - e^{-z/\kappa_j}. \tag{16}$$

The parameter κ_j is the sector-specific scale of codification resistance. Lower κ_j means that tasks in sector j are easier to standardize and reorganize into high-skill form.

Let H_t denote the stock of codified organizational knowledge. Its effective depth is

$$\phi(H_t) = \frac{H_t^{\eta}}{H_t^{\eta} + H_{1/2}^{\eta}}. \tag{17}$$

A task with hurdle z is performed in high-skill mode when the high-skill cost is below the low-skill cost. This yields the cutoff

$$z_{j,t}^{*} = \phi(H_t) \max \left\{ 1 + \log \left(\frac{p_{ls,t}}{p_{hs,t}} \right), 0 \right\}, \tag{18}$$

where

$$p_{hs,t} = \frac{w_{s,t}}{A_{hs,t} h_s}, \qquad\qquad p_{ls,t} = \frac{w_{u,t}}{A_{ls,t} h_u}. \tag{19}$$

The high-skill task share in sector j is therefore

$$\begin{aligned} x_{j,t} &= F_j(z_{j,t}^{*}) \\ &= 1 - \exp\left(- \frac{\phi(H_t) \max \left\{ 1 + \log \left(\frac{p_{ls,t}}{p_{hs,t}} \right), 0 \right\}}{\kappa_j} \right). \end{aligned} \tag{20}$$

Equation (20) is the core of the producer-service mechanism. High-skill producer-service use rises when skilled producer services become cheaper, when codified knowledge H_t expands, or when the sector's tasks are easier to codify.

The producer-service composite used by sector j is

$$PS_{j,t} = \left[x_{j,t} HS_t^{\frac{\sigma_{ps}-1}{\sigma_{ps}}} + (1 - x_{j,t}) LS_t^{\frac{\sigma_{ps}-1}{\sigma_{ps}}} \right]^{\frac{\sigma_{ps}}{\sigma_{ps}-1}}. \tag{21}$$

Appendix D shows how this two-mode representation is derived from task-level aggregation. The important implication is that sectors with lower κ_j use high-skill producer services more intensively for a given skill premium and knowledge stock.

4.4 Dynamics

The aggregate state is

$$\Omega_t = \left(L_t,\, S_t,\, \{A_{j,t}\}_{j \in \{a,m,c\}},\, H_t \right), \tag{22}$$

where L_t is population, S_t is the skilled share, $A_{j,t}$ are sectoral value-added productivities, and H_t is codified organizational knowledge.

Population evolves according to aggregate fertility:

$$L_{t+1} = n_t L_t, \qquad\qquad n_t = \sum_{i \in \{a,m,s\}} \ell_{i,t} n_{i,t}, \tag{23}$$

where $\ell_{i,t}$ is the employment share of adults of origin i.

The skilled share evolves according to the skill choices of the young:

$$S_{t+1} = p_{s,t}. \tag{24}$$

The aggregate growth rate entering the household problem is the employment-share-weighted growth rate of sectoral value-added TFP:

$$g_t = \sum_{j \in \{a,m,c\}} \ell_{j,t} \left(\frac{A_{j,t}}{A_{j,t-1}} - 1 \right). \tag{25}$$

Codified organizational knowledge evolves with education and sectoral composition. Let $\bar{e}_t$ denote average education. The economy's current codification capacity is

$$\widetilde{H}_t = \bar{e}_t \sum_{j \in \{a,m,c\}} s_{j,t} \frac{1}{\kappa_j}, \tag{26}$$

where $s_{j,t}$ is the employment share of sector j. The same κ_j that governs task assignment in (20) also governs how much sector j contributes to the expansion of codified knowledge. Sectors with lower codification resistance generate more codification capacity for a given education level and employment share.

The stock of codified knowledge evolves according to

$$H_{t+1} = H_t + \max\{\widetilde{H}_t - H_t, 0\}. \tag{27}$$

Codified organizational knowledge is irreversible: current education and sectoral composition can maintain the inherited frontier, but the frontier expands only when current codification capacity exceeds the existing stock.

Finally, sectoral productivity evolves as

$$\frac{A_{j,t+1}}{A_{j,t}} = 1 + H_t (S_t L_t)^{\varphi_j}, \qquad j \in \{a, m, c\}. \tag{28}$$

Productivity growth therefore depends on codified organizational knowledge and on the scale of skilled labor.

The mechanism is summarized by the joint evolution of sectoral employment, S_t, and H_t. Structural change reallocates labor across sectors. Because sectors differ in skill intensity and codification resistance, this reallocation changes both the demand for skilled labor and the accumulation of codified organizational knowledge. Growth feeds back into household choices by raising the obsolescence of inherited knowledge, increasing education investment and reducing fertility. In this environment, manufacturing affects long-run development by shaping the skill and knowledge base of the service economy that follows.

5 Calibration

I calibrate the model to the United States over 1850–2019. The U.S. transition provides the benchmark economy for the counterfactual exercises in Section 6: over this period the United States moves out of agriculture, experiences a pronounced manufacturing hump, shifts into services, and completes most of its demographic transition. The calibration is designed to discipline both this aggregate development path and the mechanism emphasized in the paper. The aggregate moments discipline structural change, fertility, and income growth. The mechanism moments discipline the skill content of production and, in particular, the fact that the producer-service occupations used by manufacturing are more education-intensive than the producer-service occupations used by agriculture or consumer services.

The targeted moments fall into four groups. First, I target the time-series paths of agricultural, manufacturing, and service employment shares, fertility, and normalized GDP per capita. These moments discipline the reallocation of labor across sectors, the demographic transition, and the aggregate income path. Second, I target three scalar moments that summarize the transition: the peak manufacturing employment share, average fertility, and the agricultural-to-service fertility ratio. Third, I target final-period skill moments in 2019: the aggregate skilled employment share and the skilled employment shares within value added in agriculture, manufacturing, and consumer services. These moments discipline the supply of skilled labor and the sectoral skill intensities in value-added production. Fourth, I target the producer-service moments that are most directly tied to the mechanism: the high-skill employment share within producer-service inputs used by agriculture, manufacturing, and consumer services in 2019, and the change in the aggregate high-skill producer-service share between 1960 and 2019.

The producer-service targets are central. I construct them from the 2019 IPUMS USA Census sample by first classifying workers into broad industries—agriculture, manufacturing, and consumer services—and then restricting attention within each broad industry to producer-service occupations. I then compute the share of those producer-service workers with at least one year of college. The empirical object is therefore not whether all manufacturing workers are more educated than all service workers. Rather, it is whether the producer-service occupations used by manufacturing are more education-intensive. This is the object mapped into the model's high-skill share within producer-service inputs. Appendix E describes the industry and occupation classification, the IPUMS construction, and an international validation exercise showing that the same pattern is not unique to the United States.

The calibration proceeds in two steps. I first fix a set of parameters externally, either by normalization or by reference to the literature. I then choose the remaining parameters jointly by simulated method of moments. The model is simulated at an annual frequency and aligned so that model period $t_0 = 60$ corresponds to 1850. Appendix E reports the full target vector, data sources, numerical procedure, and robustness exercises.

5.1 Externally set parameters

Table 5 reports the parameters fixed outside the simulated method of moments. Manufacturing is the numeraire, unskilled efficiency is normalized to one, and initial productivity levels are normalized to zero in logs. I set a model generation to 20 years, following the quantitative unified-growth literature and the parameterization in Lagerlöf (2006). The household block follows the unified-growth mechanism of Galor and Weil (2000), and final demand follows the non-homothetic CES structure of Comin et al. (2021). I set the elasticity of substitution across final goods to $\sigma = 0.3$, consistent with low substitutability across broad consumption categories. I keep the skill-choice and production elasticities parsimonious: the coefficient on the log skill premium is normalized to $\beta_\pi = 1$, the logit dispersion parameter is set to $\sigma_{\text{skill}} = 0.1$, and the elasticities of substitution across skill types and producer-service modes are set to 1.2, values in the range commonly used in quantitative work with skill-complementary production structures (Krusell et al., 2000; Acemoglu and Autor, 2011). Appendix E reports sensitivity to these fixed elasticities.

I also allow input-output coefficients to evolve over the transition. The economy starts in 1850 with value-added production, $\theta_{j,1850} = 1$, and input-output coefficients gradually converge to their modern values by 2025.[7] This choice is motivated by evidence that changes in U.S. industry output reflect changes in input-output coefficients as well as changes in final demand (Feldman et al., 1987). It also follows the distinction emphasized by Herrendorf et al. (2013): households consume final expenditure categories, while production is organized through value added and intermediate inputs.

5.2 Internally calibrated parameters

The remaining parameters are chosen jointly. Let

$$\Theta = \left(\gamma, \tau_a, \varphi_a, \varphi_m, \varphi_c, g^{GO}, \kappa_a, \kappa_m, \kappa_c, \eta_H, c_{col}, \alpha_a, \alpha_m, \alpha_c, \varepsilon_a, \varepsilon_c, w_a, w_c\right)$$

denote the internally calibrated vector. These parameters govern fertility preferences, sectoral fertility differences, sectoral productivity growth, gross-output productivity growth, producer-service codification, the cost of becoming skilled, sectoral skill intensities, and non-homothetic demand. Table 6 reports the calibrated values. The calibration is joint, but the table associates each parameter with the moment that most directly disciplines it.

7. I construct the modern coefficients from BEA input-output tables by mapping detailed industries into agriculture, manufacturing, consumer services, and producer services, and by computing sector-specific shares of value added, agricultural intermediates, manufacturing intermediates, and producer-service intermediates in gross output. Appendix E describes the mapping and construction in detail.

Table 5: Externally Set Parameters and Normalizations

Parameter	Value	Source / rationale
$p_{m,t}$	1	Manufacturing numeraire
h_u	1	Normalization
h_s	1.6	Fixed skill-efficiency premium
$\log A_{j,0}$	0	Initial productivity normalization
gen	20	Generation length, Lagerlöf (2006)
ε_m	1	Manufacturing income-effect normalization
w_m	3.5	Manufacturing demand-weight normalization
σ	0.3	Final-goods elasticity, Comin et al. (2021)
β_π	1	Skill-choice index normalization
σ_{skill}	0.1	Skill-supply response to the skill premium
σ_L	1.2	Skill-substitution elasticity, Krusell et al. (2000)
σ_{ps}	1.2	Elasticity across producer-service modes
τ_m, τ_c	1, 1	Child-rearing cost normalization
t_0	60	Model period aligned with 1850
$\theta_{j,1850}$	1	Initial value-added production
t_{GO}	2025	IO coefficients reach modern values

Notes: Appendix E reports the remaining fixed numerical parameters and the sensitivity of the results to the fixed elasticities.

The different groups of moments identify different parts of the model. Employment shares and final-demand moments discipline the non-homothetic demand parameters and the size of the manufacturing hump. Fertility moments discipline the household block and sectoral differences in child-rearing costs. Final-period skilled employment shares discipline the cost of becoming skilled and the skill intensities of value added. Sectoral productivity growth rates discipline the scale effects in productivity. The producer-service moments discipline κ_j, the sector-specific codification resistance parameters. These parameters are the main calibrated objects for the mechanism.

The calibration implies

$$\kappa_m < \kappa_c < \kappa_a.$$

Thus, manufacturing-related producer-service tasks are the easiest to reorganize into high-skill form, followed by producer-service tasks used by consumer services and agriculture. This ordering is not imposed directly. It is disciplined by the 2019 high-skill employment shares within producer-service inputs by final sector. In the calibrated economy, manufacturing matters not only because it employs workers directly, but because its intermediate-service needs are especially connected to high-skill organizational and technical tasks.

Table 6: Internally Calibrated Parameters

Parameter	Description	Value	Main target	Data	Model
γ	Utility weight on children	0.044	Average fertility	2.474	2.395
τ_a	Agricultural child cost	0.419	Fertility ratio A/S	2.000	1.977
φ_a	Agriculture scale effect	0.931	Agriculture TFP growth	0.037	0.040
φ_m	Manufacturing scale effect	0.828	Manufacturing TFP growth	0.031	0.030
φ_c	Services scale effect	0.528	Services TFP growth	0.014	0.012
g^{GO}	Gross-output TFP growth	0.003	GDPpc path deviation	0.000	0.145
κ_a	Codification resistance, A	1.007	HS in PS A, 2019	0.39	0.42
κ_m	Codification resistance, M	0.415	HS in PS M, 2019	0.65	0.71
κ_c	Codification resistance, C	0.731	HS in PS C, 2019	0.53	0.57
η_H	Frontier curvature	4.876	Δ aggregate HS in PS	0.182	0.168
c_{col}	Skill-acquisition cost	0.271	Skill share, 2019	0.345	0.308
α_a	Skill intensity, A	0.180	Within-VA skill A	0.160	0.161
α_m	Skill intensity, M	0.254	Within-VA skill M	0.220	0.229
α_c	Skill intensity, C	0.237	Within-VA skill C	0.200	0.213
ε_a	Agriculture income effect	0.016	Average agriculture share	0.259	0.256
ε_c	Services income effect	1.087	Average services share	0.560	0.537
w_a	Agriculture demand weight	0.952	Agriculture share, 1850	0.606	0.620
w_c	Services demand weight	4.035	Services share, 1850	0.231	0.204

Notes: A = agriculture, M = manufacturing, C = consumer services, PS = producer services, HS = high-skill. The calibration is joint; the target column reports the moment most closely associated with each parameter. Producer-service high-skill targets are constructed from the 2019 IPUMS USA Census sample by restricting to producer-service occupations within each broad industry and computing the share of workers with at least one year of college.

5.3 Estimation

Let m^{data} denote the vector of empirical moments and $m^{model}(\Theta)$ the corresponding simulated moments. I choose Θ to minimize

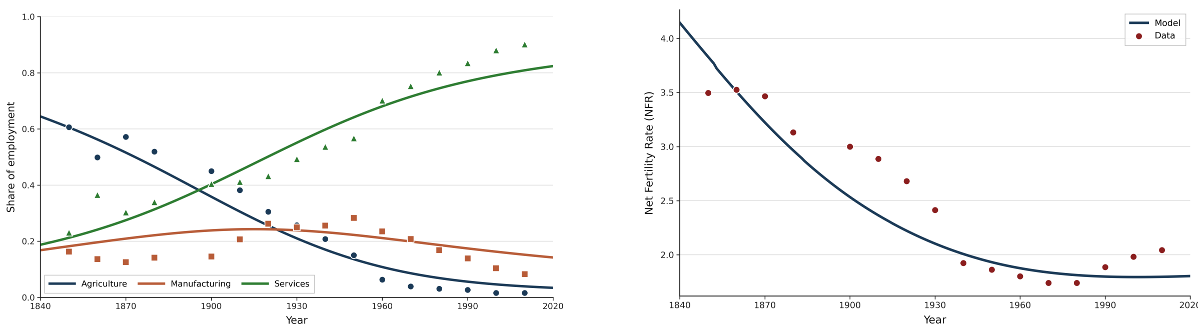
$$\mathcal{L}(\Theta) = \frac{1}{K}\sum_{k=1}^{K}\left(\frac{m_k^{model}(\Theta) - m_k^{data}}{\max\left\{\left|m_k^{data}\right|, \underline{s}_k\right\}}\right)^2 + \mathcal{P}(\Theta), \tag{29}$$

where $\underline{s}_k$ is a moment-specific scale floor. For strictly positive level variables, such as normalized GDP per capita and the agricultural-to-service fertility ratio, I use log residuals. The term $\mathcal{P}(\Theta)$ penalizes high-frequency oscillations in the skill premium and the aggregate skilled share. These penalties are not additional economic targets; they rule out parameter vectors that match isolated moments through implausibly volatile transition paths.

The minimization uses a global dual-annealing routine followed by a Powell local search. The model is overidentified: the number of targeted moments exceeds the number of internally calibrated parameters. Appendix E provides the full numerical details. In particular, the half-saturation parameter in the codification frontier, $H_{1/2}$, is normalized relative to the simulated scale of knowledge accumulation. For each candidate parameter vector, I first run a probe simulation with a fixed value of $H_{1/2}$. I then set $H_{1/2}$ equal to the average of the final observations of the resulting H_t path, subject to lower and upper bounds. This prevents the arbitrary units of H_t from mechanically determining the location of the codification frontier.

5.4 Model fit

Figure 7 compares the calibrated model to the main U.S. transition moments. The model reproduces the broad path of structural change: agricultural employment declines, manufacturing rises and then falls, and services become the dominant sector. It also captures the long-run demographic transition, with fertility declining from high nineteenth-century levels to modern low fertility. These moments discipline the two aggregate transitions that the model is built to connect: the reallocation of labor across sectors and the movement from high fertility to higher human-capital investment.



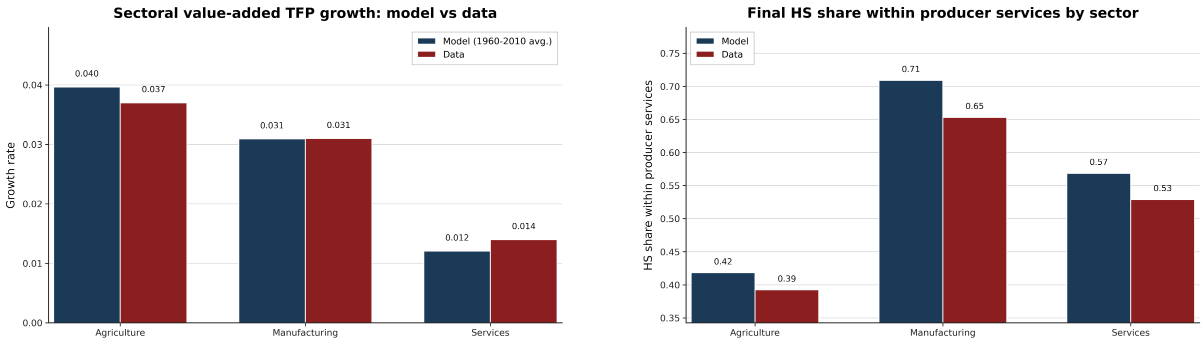
(a) Structural change

(b) Demographic transition

Figure 7: Model fit to targeted U.S. moments, 1850–2019. Dots represent data and lines show model simulations. Panel (a) shows employment shares in agriculture, manufacturing, and services. Panel (b) shows the fertility transition.

Figure 8 reports the fit for the moments most directly tied to the production side of the model. Panel (a) compares sectoral value-added productivity growth rates. The model matches the ranking of growth rates across sectors, with faster productivity growth in agriculture and manufacturing than in services. Panel (b) compares the high-skill employment share within producer-service inputs by final sector. The model reproduces the central empirical ranking: producer services used by manufacturing are the most high-skill intensive, followed by those used by consumer services and agriculture. This ranking is the empirical counterpart of the model's codification mechanism. Because the calibrated economy assigns the lowest codification resistance to manufacturing, manufacturing-related producer-service tasks are the easiest to reorganize into high-skill form.

Table 6 summarizes the fit of the main scalar targets. The model matches average fertility closely, generating an average fertility rate of 2.395 compared with 2.474 in the data, and reproduces the agricultural-to-service fertility ratio, with a model value of 1.977 compared with a target of 2.000. It also matches long-run sectoral productivity growth: agriculture, manufacturing, and services grow at 0.040, 0.030, and 0.012 in the model, compared with 0.037, 0.031, and 0.014 in the data. The model produces a final aggregate skilled share of 0.308, compared with 0.345 in the data. For the producer-service targets, the model matches the ranking across sectors and closely tracks the levels: 0.42 versus 0.39 in agriculture, 0.71 versus 0.65 in manufacturing, and 0.57 versus 0.53 in consumer services. The important discipline for the mechanism is that



(a) VA productivity growth

(b) HS shares in producer services

Figure 8: Model fit to mechanism-targeted moments. Panel (a) compares sectoral value-added productivity growth rates in the model and the data. Panel (b) compares the final high-skill employment share within producer-service inputs by final sector. Data sources: GGDC 10-sector database for value-added productivity growth and the 2019 IPUMS USA Census sample for high-skill employment shares within producer-service inputs.

manufacturing-related producer services are substantially more high-skill intensive than those used by the other final sectors, and the calibrated model reproduces this pattern.

Figure 9 reports the fit of normalized GDP per capita. GDP per capita is targeted over the calibration window, but matching this path is demanding because it is jointly shaped by sectoral productivity growth, labor reallocation, fertility, skill accumulation, and the evolution of input-output linkages. The calibrated model tracks the long-run increase in U.S. income and the acceleration of income growth over the development process.

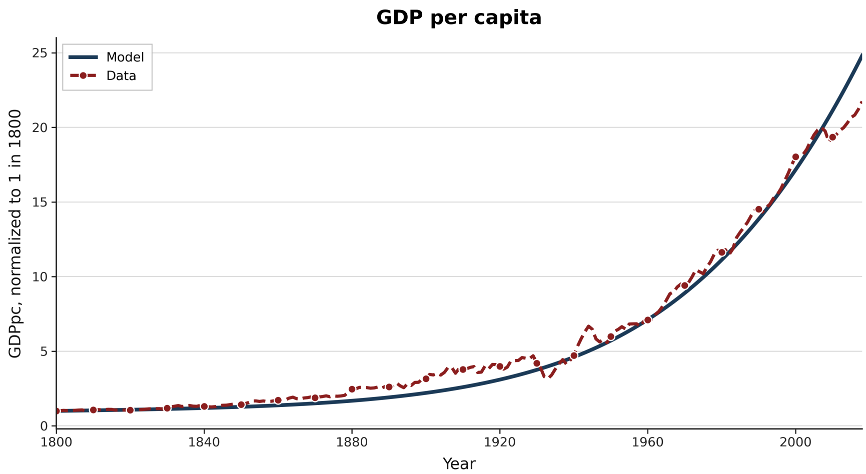
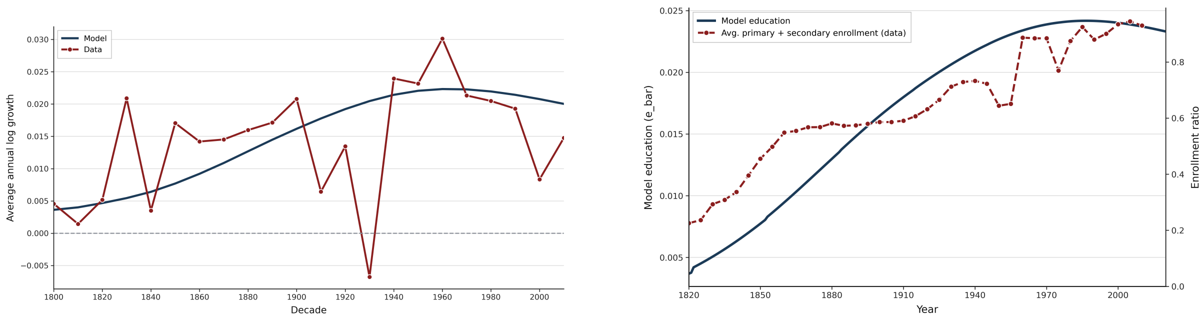


Figure 9: Model fit to normalized GDP per capita, 1850–2019. GDP per capita is normalized to one in 1850. Data source: Gapminder.

I next examine non-targeted validation moments. Figure 10 compares model-implied GDP per capita growth and schooling to their empirical counterparts. These series are not directly targeted in the calibration. The model captures the broad decline in

GDP per capita growth over development and generates a rising education path consistent with the long-run increase in schooling. In the model, schooling corresponds to the average education investment chosen by households; in the data, I compare it to average enrollment rates. This validation is informative because education responds endogenously to the growth-induced obsolescence mechanism rather than being targeted directly as a time series.



(a) GDP per capita growth

(b) Schooling

Figure 10: Non-targeted validation moments, 1800–2019. Panel (a) compares model-implied GDP per capita growth to the data. Panel (b) compares model education to enrollment rates. Schooling in the model is the average education choice across sectors; in the data it is the average enrollment rate, measured as the average of primary and secondary enrollment. Data sources: Gapminder and U.S. Censuses of Population.

As a final validation exercise, I compare the model and the data on the relationship between peak manufacturing employment and observable development outcomes. Figure 11 reports coefficients from regressions of key outcomes on the peak manufacturing employment share, estimated both in the model and in the data. These slopes are not used to estimate the U.S. calibration. The model nevertheless reproduces the qualitative pattern in the data: economies with larger manufacturing peaks have higher college attainment, larger high-skill service shares, faster GDP per capita growth, and faster demographic transitions. This exercise connects the calibrated U.S. mechanism back to the cross-country and county-level evidence in Sections 2 and 3.

The calibrated model therefore reproduces the main U.S. transition and the mechanism-specific moments needed for the counterfactual analysis. It matches the size and timing of the manufacturing phase, the demographic transition, the long-run ranking of sectoral productivity growth, and the high-skill intensity of manufacturing-related producer services. It also generates non-targeted movements in schooling and GDP per capita growth, and reproduces the qualitative cross-economy relationship between peak manufacturing and development outcomes. I now use this calibrated economy to study how changing the size of the manufacturing phase affects long-run development.

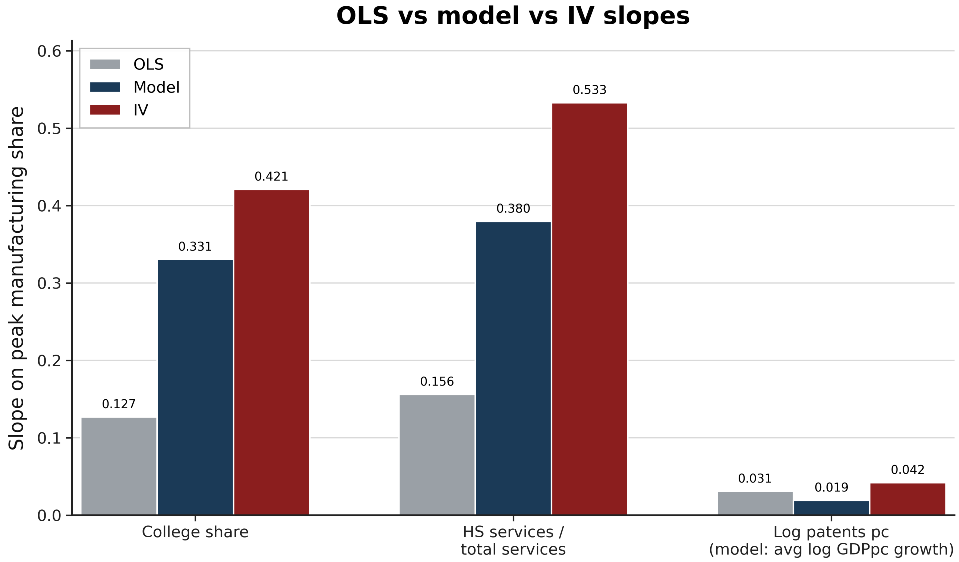


Figure 11: Non-targeted validation: model and data slopes with respect to peak manufacturing employment. The figure compares coefficients from regressions of key observable outcomes on the peak manufacturing employment share in the model and in the data.

6 Results

This section uses the calibrated U.S. economy to study how the size of the manufacturing phase affects long-run development. I generate counterfactual economies by varying the manufacturing demand weight w_m in the non-homothetic final-demand system. Varying w_m changes the relative demand for manufacturing along the transition and therefore generates economies with different peak manufacturing employment shares. I then ask how these differences translate into income, growth, education, population, and the composition of services.

The exercise is not meant to evaluate a specific policy instrument. It is a comparative-static exercise that varies the size of the temporary manufacturing phase while holding fixed the rest of the calibrated economy. Appendix D shows that the same allocations can be implemented through relative-price wedges. The policy appendix studies a more structural source of lower manufacturing exposure, through openness to trading partners with a comparative advantage in manufacturing. The purpose of the present exercise is instead to isolate the role of the manufacturing phase itself: does a smaller manufacturing peak leave an economy with a different service-sector structure and a different long-run development path?

6.1 The Manufacturing Phase in the Calibrated Economy

Figure 12 reports the main comparative statics in the calibrated economy. Outcomes are normalized to the benchmark U.S. calibration. Economies with larger manufacturing peaks reach higher GDP per capita and higher long-run growth. They also experience lower cumulative population growth.

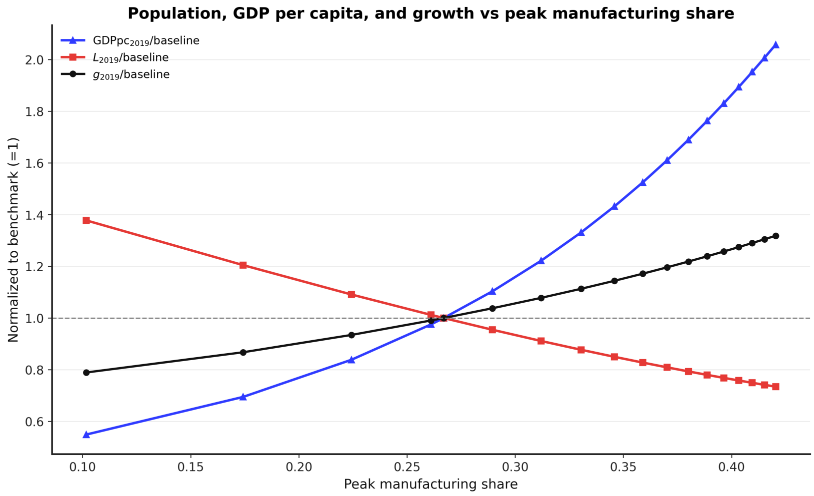


Figure 12: Comparative statics in the calibrated U.S. economy. The figure shows the relationship between the peak manufacturing employment share and GDP per capita, population, and long-run growth in 2019, with outcomes normalized to the U.S. benchmark. Counterfactual economies are generated by varying the manufacturing demand weight w_m in the final-demand system.

The mechanism is the one embedded in the model. A larger manufacturing phase raises demand for high-skill producer-service tasks and expands codified organizational knowledge. This increases the return to education, shifts household choices from child quantity toward child quality, and leaves the economy with a more skill-intensive service sector after manufacturing employment declines. Manufacturing therefore matters not because it remains large in the long run, but because it changes the knowledge and skill base of the service economy that follows.

6.2 Accounting for Cross-Country Patterns

I next ask whether this mechanism can account for the cross-country relationships documented in Section 2. For each country in the data, I compute its peak manufacturing employment share and assign it the model economy with the same peak manufacturing peak, interpolating across the simulated grid. The resulting model prediction is intentionally one-dimensional. It asks how much cross-country variation can be generated by differences in the size of the manufacturing phase alone, holding fixed all other parameters of the calibrated economy.

This distinction is important for interpreting the results. The exercise is not a full variance decomposition. The data contain measurement error and many country-specific forces that are outside the model, including institutions, geography, trade policy, natural resources, migration, demographic shocks, and education policy. I therefore do not interpret the model as trying to match the entire cross-country dispersion in outcomes. Instead, the exercise asks whether peak manufacturing acts as a sufficient statistic for a systematic component of the data.

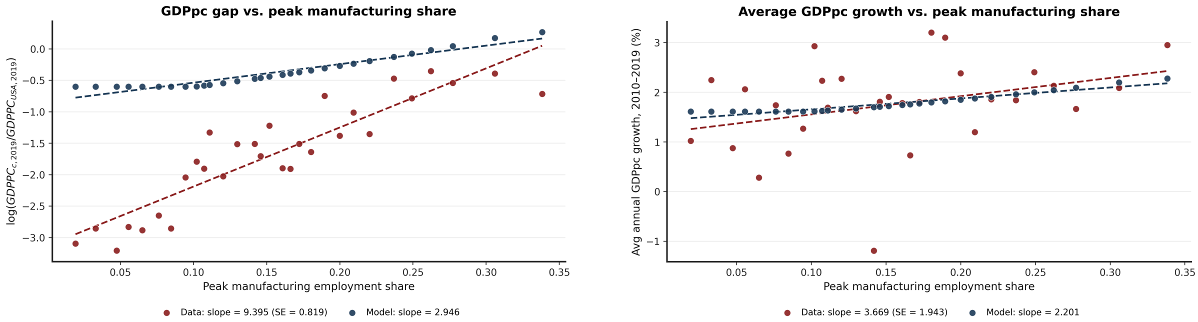
I summarize the comparison using two sets of moments. The figures report binscatter relationships between each outcome and peak manufacturing employment in the data and in the model. These figures show whether the model generates the correct empirical gradients. Table 7 reports country-level diagnostics comparing the model prediction for each country with the corresponding empirical outcome. The correlation and associated R^2 measure whether the model ranks countries correctly. The ratio of model to data standard deviations measures the amplitude of model-implied cross-country dispersion. A low standard-deviation ratio does not imply that the mechanism is absent; it means that the one-dimensional model prediction is compressed relative to the raw data.

Figure 13 reports the income results. Panel (a) plots final log GDP per capita relative to the United States against peak manufacturing employment. Countries with larger manufacturing peaks have smaller income gaps relative to the United States, and the model generates the same positive relationship. At the country level, the model-implied GDP per capita gaps have a correlation of 0.58 with the data and an associated R^2 of 0.34. Thus, a substantial component of the cross-country income ranking is aligned with the model prediction based only on peak manufacturing.

At the same time, the model-implied income differences are compressed. The standard deviation of the model-predicted GDP per capita gap is 21 percent of the empirical standard deviation. This is consistent with the interpretation of the exercise: the manufacturing mechanism accounts for a systematic development gradient, but not for the full cross-country distribution of income. The remaining dispersion reflects country-specific factors outside the model.

Panel (b) shows the corresponding relationship for average GDP per capita growth over 2010–2019. The model generates a positive relationship between peak manufacturing and subsequent growth, but the country-level fit is weak. The correlation between model and data is 0.13, and the model generates only 8 percent of the empirical standard-deviation dispersion in growth rates. I therefore interpret the growth comparison cautiously. Realized growth over a particular decade is affected by business-cycle conditions, commodity prices, financial crises, policy changes, convergence dynamics, and other shocks that are not modeled here. The model is designed to speak to long-run development paths rather than to the cross-country distribution of growth rates in one recent decade.

Figure 14 turns to the two intermediate outcomes at the center of the mechanism: service composition and education. Panel (a) shows that economies with larger manufacturing peaks have larger FIRE shares in services. In the model, this object corresponds to high-skill producer-service employment as a share of service employment. This is the outcome for which the model performs best in terms of dispersion. Table 7 shows that the model generates 87 percent of the empirical standard-deviation dispersion in FIRE shares, with a model-data correlation of 0.50 and an associated R^2 of 0.25. The model therefore not only generates the positive association between manufacturing



(a) $\log(GDPPC_{c,2019}/GDPPC_{USA,2019})$ **(b)** Average GDPpc growth

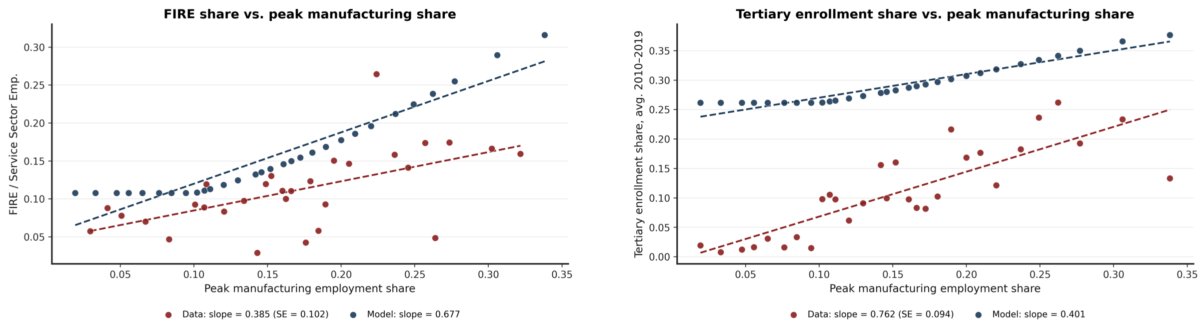
Figure 13: Peak manufacturing and income outcomes. Panel (a) plots final log GDP per capita relative to the United States against peak manufacturing employment. Panel (b) plots average GDP per capita growth over 2010–2019 against peak manufacturing employment. Red points and lines correspond to the data; blue points and lines correspond to model-generated economies.

peaks and high-skill services, but also produces a quantitatively similar amount of cross-country dispersion in this object.

This comparison should nevertheless be interpreted with some caution. In the model, high-skill producer services are a task-based object. These tasks may be performed inside manufacturing firms, inside service firms, or in specialized producer-service industries. In the data, I proxy for this object using employment in a fixed set of FIRE and professional-service industries. Since the boundary between in-house producer-service work and market producer-service industries varies across countries, the FIRE comparison is not a literal accounting of a fixed industry category. It is instead a test of whether the mechanism generates the predicted shift toward high-skill producer-service activity. On this dimension, the model performs well.

Panel (b) shows that the model also generates the positive relationship between peak manufacturing and tertiary education. The model-data correlation is 0.48, with an associated R^2 of 0.23. The standard-deviation ratio is 0.28, implying that the model generates 28 percent of the empirical dispersion in tertiary enrollment. Thus, the model captures a meaningful education gradient, but it does not reproduce the full cross-country dispersion in schooling outcomes. This is unsurprising: education levels also reflect schooling policy, public finance, institutions, demographics, and measurement differences across national education systems. The relevant result is that, although the model is calibrated to the U.S. transition, it generates the positive cross-country association between manufacturing peaks and human-capital accumulation.

Figure 15 reports the demographic implication. Economies with larger manufacturing peaks experience lower cumulative population growth. The model generates the same negative relationship as in the data. This follows from the quantity-quality channel: a larger manufacturing phase raises skill demand and growth, higher growth raises education investment, and higher education investment lowers fertility.



(a) FIRE share of services

(b) Tertiary education

Figure 14: Peak manufacturing, service composition, and education. Panel (a) plots the FIRE share of service-sector employment against peak manufacturing employment. In the model, FIRE corresponds to high-skill producer-service employment as a share of service employment. Panel (b) plots tertiary enrollment over 2010–2019 against peak manufacturing employment. Red points and lines correspond to the data; blue points and lines correspond to model-generated economies.

At the country level, the model-data correlation for population growth is 0.37, with an associated R^2 of 0.14. The standard-deviation ratio is 0.13, so the model-implied demographic differences are again compressed relative to the data. This is to be expected. Long-run population growth reflects mortality transitions, migration, disease environments, wars, and demographic policies, none of which are directly modeled here. The relevant finding is that the manufacturing mechanism generates the correct sign and a nontrivial cross-country association with the demographic transition.

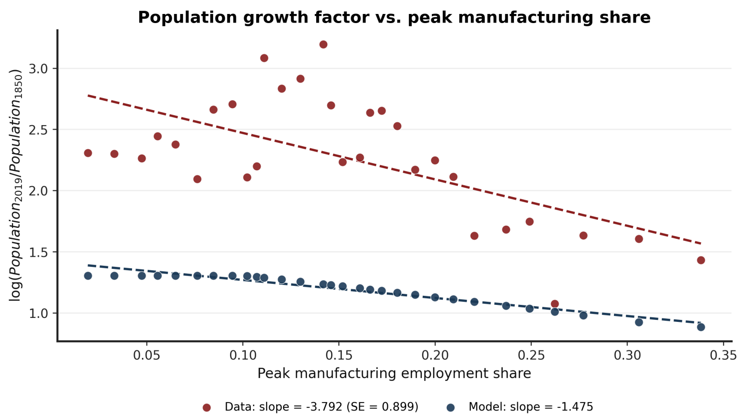


Figure 15: Peak manufacturing and population growth. The figure plots $\log(Population_{2019}/Population_{1850})$ against peak manufacturing employment. Red points and lines correspond to the data; blue points and lines correspond to model-generated economies.

Table 7 summarizes the country-level fit diagnostics. The strongest result is for FIRE intensity: the model generates nearly the same amount of cross-country dispersion as the data and has a moderate positive correlation with the empirical measure. For GDP per capita and tertiary education, the model has meaningful predictive content, with correlations of 0.58 and 0.48, respectively, but under-generates the magnitude of cross-country dispersion. For population growth, the model also predicts the correct direction, though the model-implied dispersion is again small relative to the data. For average GDP per capita growth over 2010–2019, the fit is weak.

Table 7: Country-Level Model Fit

Outcome	N	σ_m/σ_d	Corr.	R^2	Slope ratio
GDP per capita	172	0.207	0.579	0.336	0.309
Tertiary enrollment	172	0.277	0.482	0.233	0.516
GDP per capita growth	172	0.081	0.127	0.016	0.606
Population growth	172	0.130	0.368	0.135	0.403
High-skill services	51	0.870	0.498	0.248	1.618

Notes: σ_m/σ_d is the ratio of model to data standard deviations. Corr. and R^2 compare country-level model predictions with the data. The slope ratio is the model slope divided by the data slope from separate regressions on peak manufacturing employment. All countries with non-missing data are included.

The quantitative message is therefore not that the manufacturing mechanism explains all cross-country dispersion. It does not, and the exercise is not designed to do so. The model varies a single structural force, while the data reflect many country-specific histories and shocks. The result is instead that the size of the manufacturing phase generates systematic and persistent differences in the type of service economy that follows. Economies with smaller manufacturing peaks can still move into services, but they enter services with less human capital, weaker high-skill producer-service capacity, higher fertility, and lower long-run income. The model therefore rationalizes the central empirical pattern of the paper: development without industrialization is possible, but it produces a different service economy.

7 Conclusion

This paper studies the long-run consequences of developing without industrialization. Labor can move out of agriculture and into services without a large manufacturing phase, but this transition need not produce the same service economy. Manufacturing creates demand for organizational, technical, managerial, and professional tasks that are intensive in skilled labor. These tasks help build the human-capital and knowledge base of the service sector that follows.

The evidence points in this direction. Across countries, higher peak manufacturing employment shares are associated with faster growth after the transition into services, larger gains in tertiary education, faster fertility declines, and more skill-intensive service sectors. Within the United States, counties with larger historical manufacturing booms later develop more education-intensive service occupations. I interpret these facts through a unified growth model with structural change, endogenous fertility, education, and heterogeneous producer services. In the model, manufacturing raises demand for producer-service tasks that are relatively easy to codify and reorganize into high-skill form. Calibrated to the United States over 1850–2019, the model accounts for 31 percent of the relationship between peak manufacturing and relative income, 60 percent of the relationship with GDP per capita growth, 53 percent of the relationship with tertiary education, and 39 percent of the relationship with cumulative population growth.

Development without industrialization is therefore possible, but it is not neutral. What matters is not only that services expand, but what kind of services expand. A path that bypasses manufacturing is more likely to end in a service sector concentrated in lower-skill consumer-facing activities. A path with a larger manufacturing phase is more likely to leave behind the skills, institutions, and producer-service capacity that support a higher-growth service economy.

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A Data Sources and Variable Construction

This appendix describes the data sources and variable construction used in the empirical analysis. The paper combines cross-country data on income, sectoral employment, fertility, and education with U.S. county-level data on historical manufacturing activity, occupational skill content, geography, patents, education, income, and employment composition. Country-level variables are harmonized at the country-year level. Historical U.S. county-level variables are harmonized to 1850 county boundaries unless otherwise stated.

A.1 Cross-country data

A.1.1 GDP per capita

Source note. Gapminder requests the citation: "GAPMINDER: GDP per capita v25 – constant PPP$ 2011 based on World Bank, Maddison & IMF." Documentation: https://www.gapminder.org/sources/gdp-per-capita/.

GDP per capita is from Gapminder's GDP per capita series. The variable measures output per person in purchasing-power-parity adjusted international dollars. Gapminder constructs the series to make income levels comparable across countries and over time by adjusting for inflation and cross-country price differences. For recent decades, the series relies primarily on World Bank PPP GDP per capita data. For years before 1990, Gapminder uses historical income estimates from the Maddison Project and aligns these historical series to the modern PPP benchmark. For the most recent years in which World Bank observations are not yet available, Gapminder extends the series using IMF World Economic Outlook forecasts. The historical series is available back to 1800.

In the empirical analysis, I use the log of GDP per capita. Annual log GDP per capita growth is computed as

$$g_{c,t} = \log GDPpc_{c,t} - \log GDPpc_{c,t-1}.$$

For the main cross-country growth fact, the dependent variable is the average annual log GDP per capita growth rate over 2010–2019:

$$\bar{g}_{c,2010-2019} = \frac{1}{9} \sum_{t=2011}^{2019} \left(\log GDPpc_{c,t} - \log GDPpc_{c,t-1} \right).$$

I exclude 2020 from this exercise to avoid the effect of the COVID-19 shock.

A.1.2 Fertility

Source note. Fertility is from Gapminder's "babies per woman" series, corresponding to the total fertility rate. Documentation: https://www.gapminder.org/data/documentation/gd008/.

Fertility is from Gapminder's "babies per woman" series. The variable corresponds to the standard total fertility rate, defined as the average number of children a woman would have given current age-specific fertility rates. The series combines modern demographic data with historical estimates and extends back to 1800. I use the country-year total fertility rate to study the relationship between industrialization and the demographic transition.

For the cross-country fertility regression, the dependent variable is the average fertility rate over 1800–2020:

$$\overline{TFR}_c = \frac{1}{221} \sum_{t=1800}^{2020} TFR_{c,t}.$$

A.1.3 Sectoral employment shares and manufacturing peaks

Source note. Post-1990 sectoral employment shares are from Gapminder variables based on ILO modeled employment estimates. ILOSTAT documentation: `https://ilostat.ilo.org/`. Historical manufacturing employment shares are supplemented using the GGDC 10-sector database: `https://www.rug.nl/ggdc/structuralchange/`.

Sectoral employment shares in agriculture, manufacturing, and services are taken from Gapminder/ILO for the post-1990 period and supplemented with the GGDC 10-sector database where available. The three broad sectors used in the paper are agriculture, manufacturing, and services.

The GGDC 10-sector database is used to extend manufacturing employment histories for countries covered by that database. In the GGDC data, the manufacturing employment share is constructed as

$$\text{Manu. Emp. Share}_{c,t} = \frac{EMP^{MAN}_{c,t}}{EMP^{SUM}_{c,t}},$$

where $EMP^{MAN}_{c,t}$ is manufacturing employment and $EMP^{SUM}_{c,t}$ is total employment in the GGDC 10-sector database. When both Gapminder/ILO and GGDC observations are available, I use the Gapminder/ILO series for the post-1990 harmonized employment shares and the GGDC series to fill earlier manufacturing observations.

The key cross-country treatment variable is the peak manufacturing employment share:

$$\text{Manu. Peak}_c = \max_t \text{Manu. Emp. Share}_{c,t}.$$

For countries with GGDC coverage, the peak is computed over the combined GGDC and Gapminder/ILO period. For countries without GGDC coverage, it is computed over the Gapminder/ILO period. The final regression sample contains 174 countries.

A.1.4 Education

Source note. Cross-country education is from the Barro-Lee educational attainment dataset. Data and documentation: `https://barrolee.com/`. The dataset is described in Barro and Lee (2013).

Cross-country education is from the Barro-Lee educational attainment dataset. I use the tertiary education measure matched to the country-year panel. Barro and Lee provide internationally comparable educational attainment measures for a large panel of countries at five-year intervals from 1950 to 2010 (Barro and Lee, 2013). In the cross-sectional figures, I average the tertiary education measure over 2010–2019 when observations are available.

For the long-difference education exercise, I compute the change in tertiary education between the first and last years in which both education and GDP per capita are observed:

$$\Delta e_c = e_{c,T_c} - e_{c,t_c},$$

and the corresponding change in log GDP per capita:

$$\Delta \log y_c = \log GDPpc_{c,T_c} - \log GDPpc_{c,t_c}.$$

Countries are then grouped by quintiles of Manu. Peak_c.

A.1.5 Cross-country controls

The main growth regressions include controls for average GDP per capita, the service employment share, years since the manufacturing peak, region fixed effects, and income-quintile fixed effects. Income quintiles are constructed from the cross-country distribution of GDP per capita. Years since the manufacturing peak are computed as the distance between the outcome period and the year in which country c reaches Manu. Peak_c.

A.2 Input-output data and service-sector composition

Source note. The input-output exercise uses the 2015 OECD Inter-Country Input-Output table. Data and documentation: `https://www.oecd.org/sti/ind/inter-country-input-output-tables.htm`.

The input-output exercise uses the 2015 OECD Inter-Country Input-Output table. I use the 2015 small ICIO table and aggregate country-industry cells to global industries. Let Z denote the matrix of intermediate flows and Y denote gross output. I construct the technical coefficient matrix

$$A_{ij} = \frac{Z_{ij}}{Y_j},$$

and compute Leontief total requirements using

$$x_j = (I - A)^{-1} f_j,$$

where f_j is a final-demand shock to sector j.

The exercise considers a \$100 final-demand shock to each of three broad using sectors: agriculture, manufacturing, and consumer services. Agriculture includes sectors $A01$, $A02$, and $A03$. Manufacturing includes all sectors whose OECD industry code begins with C. Agriculture and manufacturing shocks are allocated across detailed industries in proportion to gross output.

The market-service universe and high-skill-service classification are reported in Table 8. High-skill services are computer and information services, finance and insurance, and professional, scientific, and technical activities. All remaining market services are classified as other services.

Table 8: ICIO service-sector classification

Category	OECD ICIO industries
Market services	G, $H49$, $H50$, $H51$, $H52$, $H53$, I, $J58T60$, $J61$, $J62\_63$, K, L, M, N, R, S
High-skill services	$J62\_63$, K, M

Notes: $J62\_63$ denotes computer and information services, K denotes finance and insurance, and M denotes professional, scientific, and technical activities.

In the baseline exercise, consumer services are market-service industries for which final consumption exceeds 50 percent of gross output:

$$\frac{FinalConsumption_s}{GrossOutput_s} > 0.5.$$

Final consumption is measured using household final consumption expenditure, non-profit institutions serving households, and government final consumption columns in the ICIO table. The consumer-service shock is allocated across selected industries using observed final-consumption weights. In the robustness exercise, I instead allocate the consumer-services shock across all market services using observed final-consumption weights.

For each shock, I compute the induced service output and report the share of induced service output accounted for by high-skill services:

$$HSShare_j = \frac{\sum_{s \in HS} x_{j,s}}{\sum_{s \in Services} x_{j,s}}.$$

This is the object reported in the input-output figures. The purpose of the exercise is to compare the skill intensity of the service output induced by final demand in agriculture, manufacturing, and consumer services.

A.3 Historical U.S. county data

A.3.1 County harmonization

Source note. Historical county boundary shapefiles are from NHGIS/IPUMS. NHGIS documentation and data access: https://www.nhgis.org/.

All historical county-level variables are harmonized to 1850 county boundaries. I construct geographic crosswalks using county shapefiles for each census decade and the 1850 county shapefile. For each source-year county, I compute the area overlap with 1850 counties using an equal-area projection and assign the source county to the 1850 county with the largest overlap. The resulting crosswalk maps decade-specific county identifiers to 1850 county identifiers. For variables observed on 2000 county boundaries, I allocate counts to 1850 counties using area-weighted overlays.

A.3.2 Manufacturing peaks

Source note. County manufacturing data come from the Haines historical county dataset distributed by ICPSR, study 02896: https://www.icpsr.umich.edu/web/ICPSR/studies/02896/staff.

County manufacturing data come from historical Census of Manufactures county tabulations compiled in the Haines county dataset. I use the 1880–1940 censuses to construct the manufacturing peak because these are the most reliable county-level manufacturing tabulations. The 1910 Census of Manufactures did not provide a complete county tabulation; it tabulated cities instead, so 1910 is omitted from the county manufacturing panel.

The county manufacturing measure is the number of manufacturing workers divided by total population:

$$\text{Manu. Emp. Share}_{c,t} = \frac{\text{Manu. Workers}_{c,t}}{\text{Population}_{c,t}}.$$

The county manufacturing peak is

$$\text{Manu. Peak}_c = \max_{t \in \{1880,1890,1900,1920,1930,1940\}} \text{Manu. Emp. Share}_{c,t}.$$

If a county reaches the maximum in multiple years, I assign the first peak year:

$$T_c^{peak} = \min \left\{ t : \text{Manu. Emp. Share}_{c,t} = \text{Manu. Peak}_c \right\}.$$

A.3.3 Historical census sample and occupational skill content

Source note. Individual-level historical census data are from IPUMS USA full-count U.S. Census files. Data and documentation: https://usa.ipums.org/usa/. Occupational education scores use the 1950 IPUMS occupation-based education score variable.

The historical event-study analysis uses full-count U.S. Census data from 1850 to 1940. The baseline sample consists of fathers of children between ages 6 and 18 with non-missing county, industry, and employment information. This sample is used to construct county-year measures of school enrollment, fertility, sectoral employment, and occupational skill content.

I map industries into broad sectors using the IPUMS 1950 industry classification. Manufacturing is defined as industries with codes between 306 and 499. Services are defined as industries with codes between 506 and 946. Agriculture is the omitted primary-sector category.

Individual education is not observed in the pre-1950 full-count censuses. I therefore measure the skill content of occupations using the IPUMS occupation education score from the 1950 Census. For each occupation, this score measures the share of workers in that occupation in 1950 who had completed at least some college. I assign this fixed 1950 occupation score to workers in earlier census years and average it within county-year-sector cells. The main outcome in the county event study is the mean education score of service-sector occupations:

$$EDSCOR50^{serv}_{c,t} = \frac{1}{N^{serv}_{c,t}} \sum_{i \in (c,t,serv)} EDSCOR50_{occ(i)}.$$

This variable measures whether a county's service sector is composed of occupations that later appear as high-education occupations in the 1950 Census.

Treatment groups are based on the cross-county distribution of Manu. Peak$_c$. Event time is measured relative to T^{peak}_c. The event-study specification is estimated using the Borusyak, Jaravel, and Spiess imputation estimator (Borusyak et al., 2024). The implementation uses county and year fixed effects and the default standard-error procedure in `did_imputation`.

A.4 Geographic characteristics

Source note. Coal-field locations are from the NHGIS/USGS coal-field shapefile. Physiographic provinces are from the USGS physiographic provinces shapefile. River and lake geographic files are from the GIS files used in the county-level construction.

The geographic characteristics used in the IV-style exercises are measured from 1850 county centroids. Distances are measured in kilometers and transformed as $\log(1 + d)$. Distance to coal is the distance from the 1850 county centroid to the nearest coal-field point. Coal-field locations come from the USGS/NHGIS coal-field shapefile, restricted to points within the 1850 county coverage. Distance to the fall line is computed from the boundary between the Piedmont and Atlantic Plain physiographic regions in the USGS physiographic provinces shapefile. I also use distance to the Ohio River and an indicator for whether the county is adjacent to the Great Lakes. These variables are merged into the county panel and interacted with the manufacturing peak decade in the instrument construction described in the main text.

A.5 Long-run county outcomes from NHGIS

Source note. Long-run county outcomes are constructed from 2000 Census county files downloaded from NHGIS. The relevant local extract files are `population_nhgis0030_ts_nominal_county.csv`, `nhgis0031_ds151_2000_county.csv`, `nhgis0032_ds153_2000_county.csv`, `nhgis0036_ts_nominal_county.csv`, and `nhgis0041_ds142_2000_county.csv`. NHGIS documentation: https://www.nhgis.org/.

I construct long-run county outcomes using 2000 Census data from NHGIS and harmonize them to 1850 county boundaries using area-weighted overlays. Count variables are allocated by the area share of each 2000 county that overlaps an 1850 county. Mean income is first converted to an income sum by multiplying mean income by population, then allocated by area share and divided by allocated population.

College attainment is the fraction of the population age 25 and older with a bachelor's degree or graduate/professional degree. School enrollment is constructed from NHGIS counts of enrolled and not-enrolled school-age individuals. Per-capita income is constructed by allocating total income, defined as mean income times population, to 1850 county boundaries and dividing by allocated population.

Employment shares in agriculture, manufacturing, and services are constructed from 2000 NHGIS industry counts. I use the broad industry-by-sex counts from the 2000 Census. Agriculture combines employment in agriculture-related industries. Manufacturing combines manufacturing employment. Services combine transportation, information, finance, professional/business services, education/health, arts/accommodation, and other service categories.

For the long-run composition outcomes in Table 4, I also use the detailed 2000 NHGIS industry file `nhgis0041_ds142_2000_county.csv`. Table 9 reports the definitions of high-skill services, total services, innovation activities, advanced manufacturing, and legacy routine manufacturing.

Table 9: Long-run industry classifications

Variable	Included activities
High-skill services	Information; finance and insurance; real estate; professional, scientific, and technical services; management of companies and enterprises.
Total services	Information; finance and insurance; real estate; professional, scientific, and technical services; management; administrative support; education; health; arts; accommodation and food; other services.
Innovation activities	Data processing and information services; architectural and engineering services; computer systems design; management, scientific, and technical consulting; scientific research and development; computer and electronic products manufacturing; pharmaceutical manufacturing.
Advanced manufacturing	Computer and electronic products manufacturing; pharmaceutical manufacturing.
Legacy routine manufacturing	Textile mills; textile product mills; apparel; leather; wood; paper; printing; primary metals; fabricated metals; furniture manufacturing.

The high-skill services share is high-skill service employment divided by total service employment. The innovation employment share is innovation employment divided by total employment. The advanced manufacturing share is advanced manufacturing employment divided by total manufacturing employment.

A.6 Patents

Source note. Patent data are from PatentsView bulk data files. Data and documentation: https://patentsview.org/download/data-download-tables. The files used are `g_patent.tsv`, `g_ipc_at_issue.tsv`, `g_assignee_disambiguated.tsv`, and `g_location_disambiguated.tsv`.

Patent counts are constructed from PatentsView bulk data. I classify patents into broad technology sectors using IPC codes at issue. I construct the IPC subclass by combining IPC section, class, and subclass. Patents with an IPC subclass beginning with *A*01 are classified as agriculture. Patents with IPC subclasses beginning with *G*06, *G*07, *G*08, or *H*04 are classified as services, corresponding primarily to computing, data processing, control, and communications technologies. All remaining patents are classified as manufacturing. If a patent appears in multiple technology groups, I assign sectors using the priority order agriculture, services, and then manufacturing.

I then merge patents to assignee locations. PatentsView assignee records are linked to disambiguated location records using `location_id`. I keep assignee locations with non-missing geographic information and merge them to the patent table using `patent_id`.

Each patent-assignee location contributes one patent count. I then aggregate patent counts by county FIPS, state FIPS, and broad technology sector.

To merge patents to the historical county panel, I use the longitude and latitude attached to the PatentsView assignee location records. Patent-location observations are converted to point geometries and spatially joined to 1850 county polygons. Patent counts are then summed within 1850 counties:

$$\text{Patents}_c^k = \sum_{p \in c} \mathbb{1}\{sector(p) = k\}, \qquad k \in \{A, M, S\}.$$

The main long-run outcome in Table 4 is log patents per capita, constructed using total patent counts and the 2000 population harmonized to 1850 county boundaries.

A.7 University openings

Source note. University and college openings are collected from Wikidata and supplemented with information from English Wikipedia pages. Wikidata query service: https://query.wikidata.org/.

I collect university and college openings from Wikidata. The query retrieves institutions with a founding year, coordinates, and an English Wikipedia page. I keep institutions with non-missing founding years and locations. Institutions are assigned to counties using their coordinates. For descriptive exercises, I group counties by their historical manufacturing peak and compute the distribution of university openings across peak-manufacturing groups.

I also classify institutions by topic using the first 250 words of their Wikipedia page. Topic categories include clergy, agriculture, liberal arts, commerce, business, law, mechanical arts, medicine, military, teacher training, and other.

B Additional Cross-Country Evidence

This appendix reports additional cross-country evidence supporting the facts documented in Section 2. The exercises below are not used in the calibration. They show that the relationship between peak manufacturing employment and development outcomes is robust to alternative specifications and that the input-output mechanism is not driven by the baseline definition of consumer services. Variable definitions and data sources are described in Appendix A.

B.1 GDP per capita growth without service-employment controls

The baseline growth regressions in Section 2 control directly for the average service employment share over 2010–2019. This is a demanding specification because the service share is itself an outcome of structural change and is mechanically related to the decline of agriculture and manufacturing. Table 10 reports the same set of regressions after omitting the service-employment-share control.

Table 10: Dep. Var. is Avg GDPPC growth rate 2010-2020

	(1)	(2)	(3)	(4)
Peak Emp. % Man.	3.334*	5.164**	4.201	5.002*
	(1.83)	(2.10)	(1.54)	(1.71)
N	174	174	174	174
R2	0.0143	0.0495	0.105	0.130
Income Quintile FE	no	yes	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The coefficient on Manu. Peak$_c$ remains positive and is only marginally insignificant at the 10% level when controlling only for income quintile and regional fixed effects. Thus, the positive association between peak manufacturing employment and subsequent GDP per capita growth is not driven by directly conditioning on the size of the service sector. Instead, countries with larger manufacturing peaks grow faster even when the specification compares them without holding fixed the contemporaneous service employment share; that is, countries with a higher peak tend to grow faster even when compared to countries which are at an earlier stage of the development process which, ceteris paribus, would enjoy the advantages of catch up growth and grow faster than countries at the frontier.

B.2 Human-capital quality

Section 2 shows that countries with larger manufacturing peaks accumulate more tertiary education over development. This subsection reports a complementary exercise using human-capital quality, measured by test scores. The dependent variable is human-capital quality in 2015. The regressions control for average log GDP per capita over 2010–2019 and progressively add income-quintile fixed effects, region fixed effects, and additional controls.

Across specifications, Manu. Peak$_c$ is positively associated with human-capital quality. The relationship remains statistically significant after controlling for income levels, income-quintile fixed effects, regional fixed effects, and additional controls. This suggests that the link between manufacturing and human capital is not only about years of schooling or tertiary enrollment. Countries with larger manufacturing peaks also tend to have higher measured education quality.

Table 11: Dep. Var. is Human Capital Quality, 2015

	(1)	(2)	(3)	(4)
Avg. log GDP per capita, 2010–2019	0.618***	0.523***	0.571*	0.462
	(5.36)	(2.74)	(1.82)	(1.42)
Peak Emp. % Man.	4.943***	4.866***	3.554***	3.237***
	(3.97)	(4.05)	(2.91)	(2.66)
N	79	79	79	79
R2	0.552	0.559	0.697	0.702
Income Quintile FE	no	no	yes	yes
Region FE	no	no	yes	yes
Controls	no	no	no	yes

t statistics in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

B.3 Input-output robustness: alternative consumer-service demand

The baseline input-output exercise in Section 2 defines consumer services as market-service industries for which final consumption exceeds 50 percent of gross output. This subsection reports a robustness exercise using a broader definition. Instead of selecting only consumer-oriented service industries, I allocate the consumer-services final-demand shock across all market services using observed final-consumption weights. This allows final consumers to purchase services that are also heavily used as intermediate inputs, such as finance, computer and information services, and professional services.

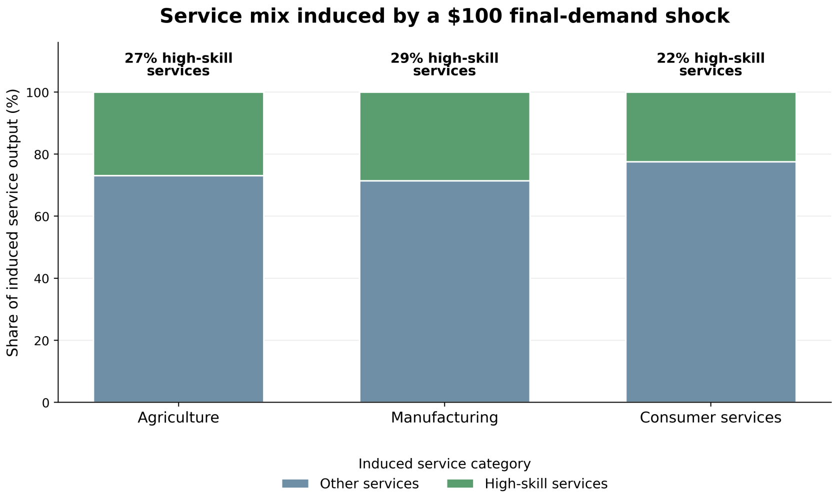

Figure 16: Robustness of the input-output service-mix exercise. Bars show the service-sector mix induced by a $100 final-demand increase in agriculture, manufacturing, or consumer services, computed using Leontief total requirements from the 2015 OECD ICIO table. In this robustness exercise, the consumer-services shock is allocated across all market services using observed final-consumption weights.

The qualitative pattern is unchanged. Manufacturing induces a more high-skill-intensive service mix than consumer-service demand, even when consumer-service demand is allowed to include high-skill market services. This supports the interpretation that manufacturing matters not only through direct manufacturing employment, but also through its intermediate-input linkages to high-skill producer services.

B.4 Input-output robustness by country income group

Figure 17 repeats the input-output service-mix exercise by country income group. This checks whether the baseline pattern is driven by a small set of high-income economies or whether manufacturing-intensive demand is associated with a more high-skill service mix more broadly.

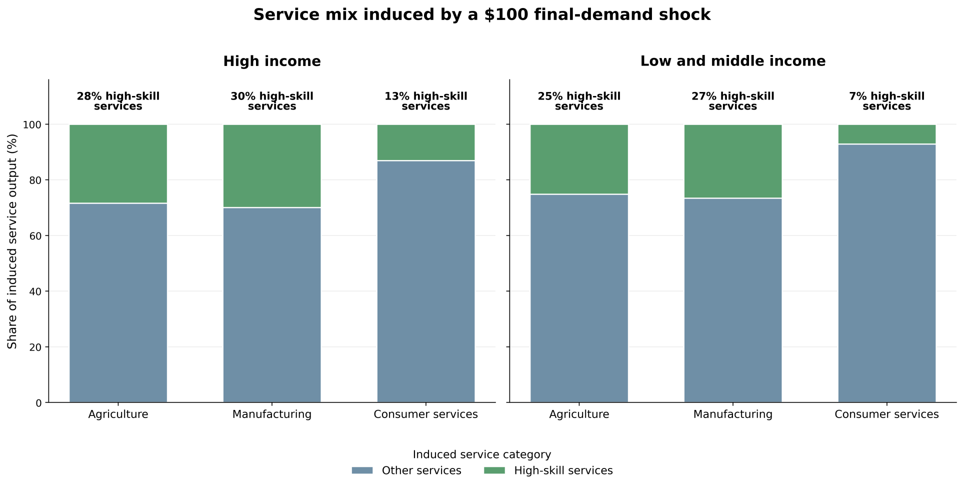

Figure 17: Input-output service-mix exercise by income group. Bars show the service-sector mix induced by a $100 final-demand increase in agriculture, manufacturing, or consumer services, computed from Leontief total requirements using the 2015 OECD ICIO table. Consumer services are defined as service industries for which final consumption exceeds 50 percent of gross output.

The manufacturing shock remains relatively intensive in high-skill services across income groups. The magnitude varies across groups, but the broad ordering is consistent with the mechanism emphasized in the paper: manufacturing demand pulls on producer-service activities that are more skill intensive than the services generated by consumer-facing final demand.

C Additional U.S. County Evidence

This appendix provides additional evidence for the U.S. county analysis in Section 3. I first document the timing and distribution of historical manufacturing peaks across counties. I then report a robustness exercise that splits counties into finer bins of the manufacturing-peak distribution. I next decompose the service-sector education-score result into the occupations that drive it. Finally, I provide complementary evidence from university openings and historical help-wanted advertisements.

C.1 Timing and size of county manufacturing peaks

Figure 18 summarizes the timing and size of county manufacturing peaks. For each county, I define Manu. Peak$_c$ as the maximum manufacturing employment share observed in the Census of Manufactures county tabulations over 1880, 1890, 1900, 1920, 1930, and 1940. The 1910 Census of Manufactures is omitted because it does not provide complete county-level manufacturing labor tabulations. The figure reports, by peak decade, the average peak manufacturing employment share among counties peaking in that decade and the number of counties reaching their peak in that decade.

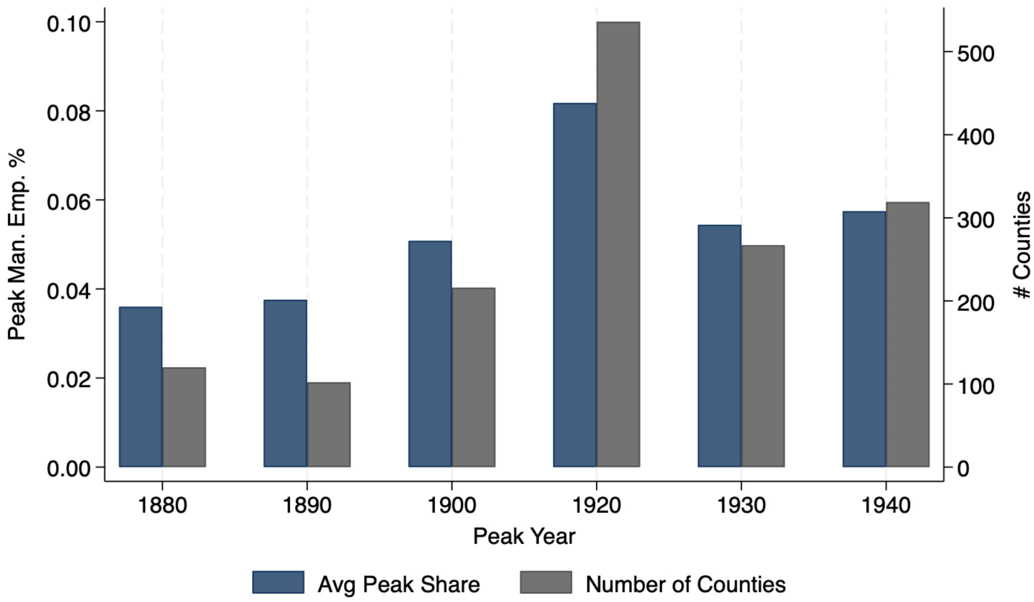

Figure 18: Timing and size of county manufacturing peaks. The figure reports the average value of Manu. Peak$_c$ among counties peaking in each decade and the number of counties reaching their manufacturing peak in that decade. County manufacturing peaks are constructed from Census of Manufactures county tabulations over 1880, 1890, 1900, 1920, 1930, and 1940.

The distribution shows substantial variation in the timing of industrialization across counties. Some counties reach their manufacturing peak during the late nineteenth century, while others peak only in the interwar period. This variation is useful for the event-study design because treatment timing differs across counties while outcomes are observed in common census decades.

C.2 Panel evidence during the U.S. transition

Before turning to the event-study design, I examine whether manufacturing-intensive counties experience different changes in fertility, schooling, and the skill content of services during the U.S. transition from a high fertility/agrarian/low schooling economy to the highly educated and industrialized country it was by the mid 20$^{\text{th}}$ century. This exercise uses the full-count Census sample from 1850 to 1940 and estimates panel regressions of the form

$$\Delta y_{ct} = \alpha + \beta \, \text{Manu. Emp. Share}_{c,t-1} + \Gamma' X_{c,t-1} + \varepsilon_{ct},$$

where y_{ct} is fertility, school enrollment, or the mean education score of service occupations in county c and decade t. The dependent variables are measured in first differences, and the main regressor is the lagged county manufacturing employment share. The controls include the urbanization rate, log population, and the agricultural employment share. The coefficient β therefore asks whether, conditional on broad measures of county development, more manufacturing-intensive counties subsequently experience larger or smaller changes in the outcome. In the more demanding specifications, I also include state-by-year fixed effects to absorb state-level shocks and policy changes, including changes in compulsory schooling laws.

Table 12: Panel Data (1850–1940)

	(1) Δ Fertility	(2) Δ Enrollment	(3) Δ Ed. Score	(4) Δ Fertility	(5) Δ Enrollment	(6) Δ Ed. Score
L.% Man Emp.	-0.282***	0.109***	6.267***	-0.312***	0.086***	2.181***
	(0.067)	(0.035)	(0.504)	(0.064)	(0.025)	(0.567)
State × Year FE	No	No	No	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	12758	12758	12748	12758	12758	12748
Within R2	0.022	0.060	0.075	0.522	0.546	0.568

Standard errors are clustered at the county level. Dependent variables are measured in first differences. Data source: US Census of Population 1850-1940.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12 reports the results. Counties with higher lagged manufacturing employment shares subsequently experience larger fertility declines, larger increases in school enrollment, and larger increases in the education score of service occupations. The signs are stable across specifications with and without state-by-year fixed effects. The magnitudes decline once state-by-year fixed effects are included, but the relationships remain statistically significant. This evidence is descriptive, but it shows that the three outcomes emphasized in the paper—fertility decline, schooling, and the skill content of services—move together with manufacturing intensity within U.S. counties during the transition.

C.3 Alternative peak-manufacturing bins

The baseline event study in Figure 5 compares mutually exclusive groups of counties in the tails of the manufacturing-peak distribution. Figure 19 reports a finer grouping. Counties are divided into seven mutually exclusive bins: the bottom decile, the

10th–20th percentiles, the 20th–30th percentiles, the middle 30th–70th percentiles, the 70th–80th percentiles, the 80th–90th percentiles, and the top decile of the Manu. Peak$_c$ distribution.

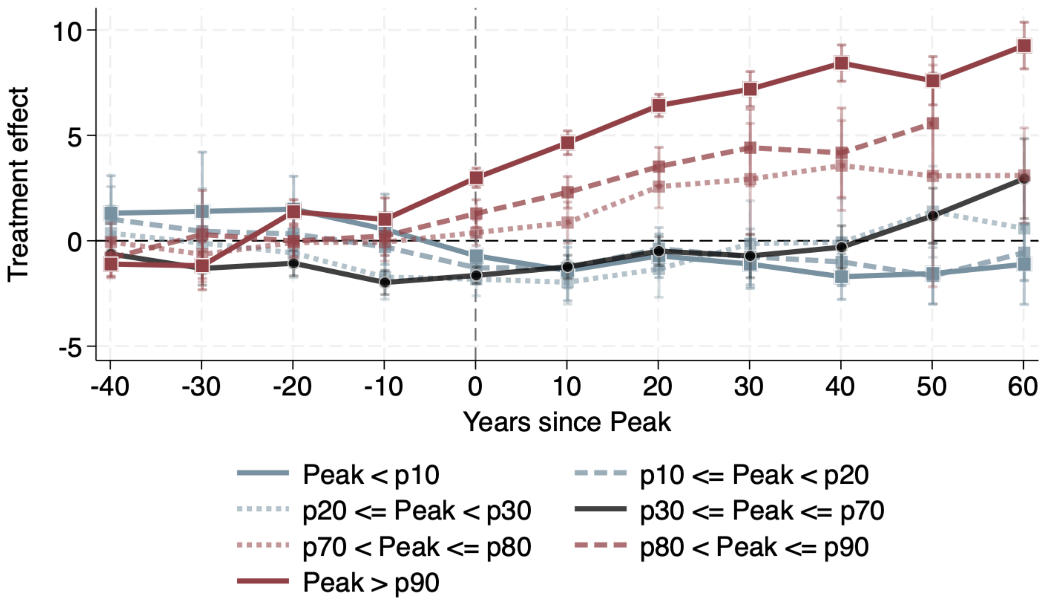

Figure 19: Event-study estimates by mutually exclusive peak-manufacturing bins. The outcome is the mean 1950 occupation education score among service-sector workers in county c and decade t. Counties are grouped by their position in the cross-county distribution of Manu. Peak$_c$. Event time is measured in years relative to the county manufacturing peak. Estimates are obtained using the Borusyak, Jaravel, and Spiess imputation estimator with county and census-decade fixed effects.

The results confirm that the baseline pattern is concentrated in the upper tail of the manufacturing-peak distribution. Counties with the largest manufacturing peaks experience the strongest post-peak increase in the educational content of service occupations. Counties in the middle of the distribution display much smaller changes, while counties in the lower tail do not exhibit a comparable increase. This pattern supports the interpretation that the intensity of the manufacturing phase matters for the subsequent composition of the service economy.

C.4 Which occupations drive the service-sector education-score result?

The event-study outcome $EDSCOR50^{serv}_{c,t}$ increases when service employment shifts toward occupations that had higher college intensity in the 1950 Census. Figure 20 decomposes this change by occupation. I compare counties in the top and bottom deciles of the Manu. Peak$_c$ distribution and compute, for each occupation, the difference-in-differences in employment-share changes between 1850 and 1950. The blue bars report this occupation-level difference-in-differences in percentage points. The orange bars report the occupation's 1950 education score.

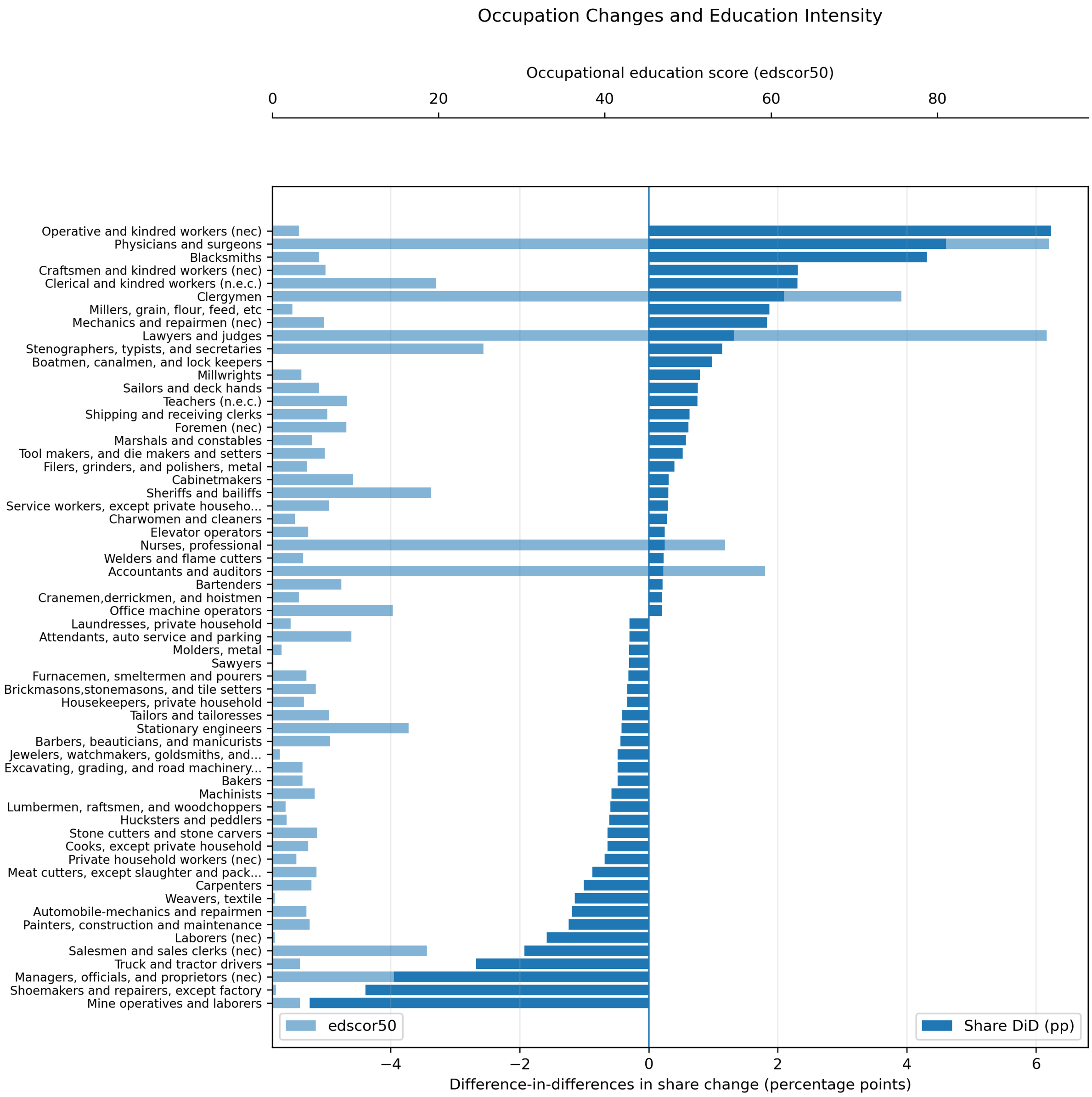

Figure 20: Occupation-level decomposition of changes in service-sector education intensity. Blue bars report the difference-in-differences in occupation employment-share changes between high- and low-peak manufacturing counties from 1850 to 1950. Orange bars report the 1950 occupation education score, defined as the share of workers in the occupation with at least one year of college. Farm occupations are excluded.

The occupations with the largest positive differential growth in high-peak manufacturing counties include physicians and surgeons, clerical workers, stenographers and secretaries, lawyers and judges, teachers, accountants, office-machine operators, and other professional or white-collar occupations. These occupations are not manufacturing production jobs, but many are organizationally complementary to manufacturing expansion: they involve accounting, administration, legal services, communication, technical support, record keeping, and coordination. By contrast, occupations with negative differential growth include laborers, salesmen, shoemakers, operatives, and other lower-education manual or routine occupations. The decomposition therefore suggests that historical manufacturing peaks are associated with a later shift in services toward the kinds of professional and organizational occupations emphasized by the mechanism.

C.5 Service industries and occupations by education intensity

Tables 13 and 14 provide additional detail on the service industries and occupations with low and high educational content in the 1950 Census. These tables help interpret the education-score outcome by showing the types of activities that receive high and low $EDSCOR50$ values.

Table 13: Service sub-industries by college intensity in 1950

Low college intensity		**High college intensity**	
Sub-industry	%	Sub-industry	%
Private households	3.5	Educational services	75.8
Shoe repair	4.3	Legal services	73.5
Trucking	5.3	Medical, excl. hospitals	71.2
Taxicabs	5.7	Welfare and religious services	59.1
Dressmaking	6.1	Engineering and architecture	58.4
Rail and bus transit	6.4	Misc. professional services	53.9
Misc. repair services	7.1	Drug stores	45.7
Sanitary services	7.2	Radio and television	43.0
Auto repair	8.0	Accounting and auditing	42.7
Water transport	8.1	Hospitals	38.5

Notes: Entries report the fraction of workers in each service sub-industry with at least one year of college in the 1950 Census.

Table 14: Service occupations by college intensity in 1950

Low college intensity		**High college intensity**	
Occupation	%	Occupation	%
Teamsters	0.0	Mathematicians	100.0
Boatmen, canalmen, lock keepers	0.0	Biological scientists	96.0
Bootblacks	0.0	Chiropractors	93.9
Telegraph messengers	0.0	Professors and instructors, n.e.c.	93.8
Midwives	0.0	Professors: engineering	93.8
Hucksters and peddlers	1.7	Professors: chemistry	93.8
Longshoremen and stevedores	2.1	Professors: biological sciences	93.8
Laundresses, private household	2.2	Professors: mathematics	93.8
Charwomen and cleaners	2.7	Professors: agricultural sciences	93.8
Private household workers, n.e.c.	2.9	Professors: physics	93.8

Notes: Entries report the fraction of workers in each occupation with at least one year of college in the 1950 Census. Manufacturing-heavy occupations are excluded from the low-college list.

The high-education service activities are concentrated in legal, medical, educational, accounting, engineering, architectural, professional, and technical occupations. This is consistent with the mechanism in the paper: manufacturing exposure predicts the later emergence of service activities that are complementary to organization, coordination, technical knowledge, and professional expertise.

C.6 University openings

I next examine whether counties with larger historical manufacturing peaks were also more likely to develop local higher-education institutions. University and college openings are collected from Wikidata, assigned to counties using institution coordinates, and grouped by founding year. Figure 21 plots the fraction of universities founded between 1860 and 2000 across counties grouped into quartiles of historical peak manufacturing intensity.

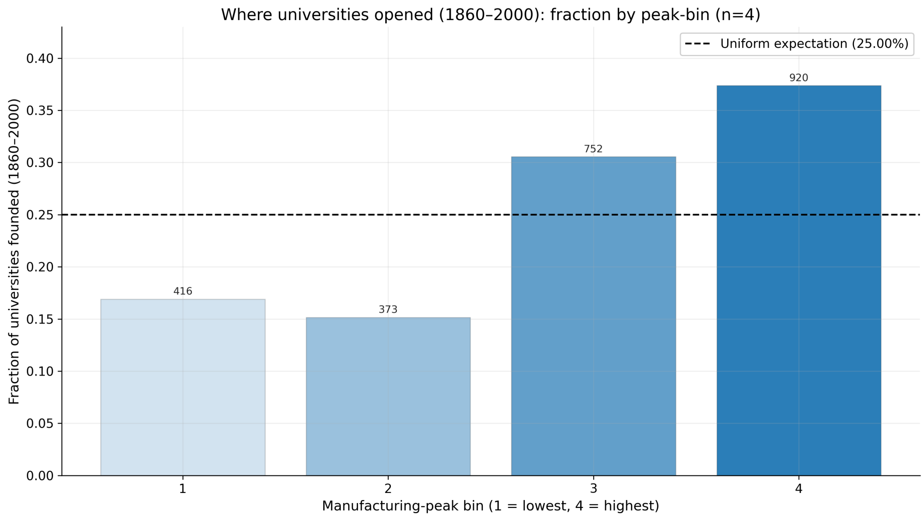

Figure 21: University openings and historical manufacturing exposure. The figure reports the fraction of universities founded between 1860 and 2000 across counties grouped into quartiles of Manu. Peak$_c$. University openings are collected from Wikidata and assigned to counties using institution coordinates.

Counties with larger manufacturing peaks account for a disproportionate share of university openings. This evidence is descriptive, but it is consistent with the idea that industrialization is associated with the local development of institutions that support skill formation and human-capital accumulation.

C.7 Help-wanted advertisements and demand for skilled labor

I also use historical help-wanted advertisements to study whether manufacturing expansion is associated with demand for skilled labor. The data are constructed from the Library of Congress *Chronicling America* newspaper repository. The underlying corpus begins from pages containing a proximity match for "help wanted." Newspapers are geolocated and grouped into geographic regions. Region-year outcomes are constructed from occupation keyword counts.

I distinguish between skilled manufacturing occupations and skilled service or white-collar occupations. Skilled manufacturing keywords include occupations such as machinist, millwright, and boilermaker. Skilled service and white-collar keywords include occupations such as clerk, bookkeeper, and stenographer. To normalize for changes in newspaper coverage and general labor-market advertising intensity, I compare these keyword counts to a baseline basket of common lower-skill occupations: cook, waiter, janitor, and laborer.

I estimate an event study around the appearance of new or expanding factory mentions in local newspapers. A region is treated in the first year in which factory-expansion mentions exceed the threshold used in the text-classification procedure. Figure 22 reports the long-run event-study estimates. The dependent variable is the log of occupation keyword hits normalized by the baseline basket of lower-skill occupations.

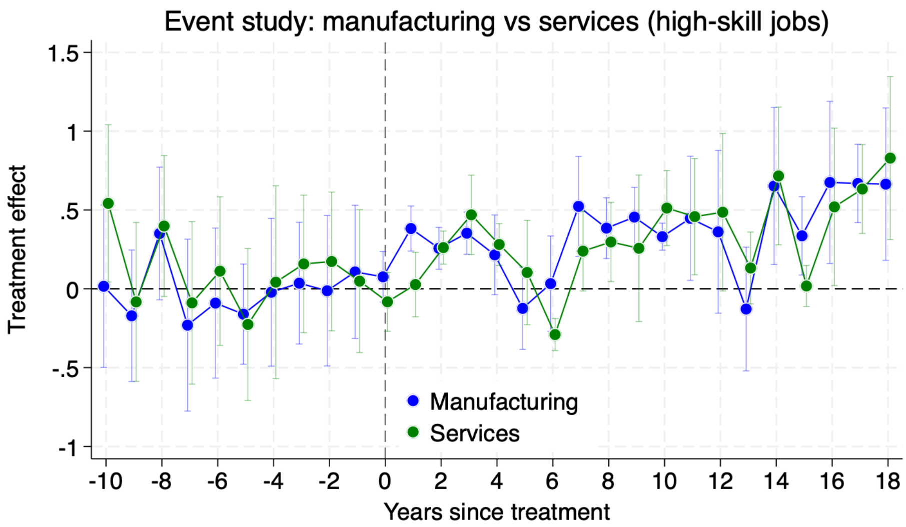

Figure 22: Long-run event-study effect of new factory mentions on demand for skilled labor. The dependent variable is the log of occupation keyword hits normalized by a baseline basket of lower-skill occupations. Estimates are obtained using the Borusyak, Jaravel, and Spiess imputation estimator on region-year newspaper data. Vertical bars report 95 percent confidence intervals clustered by region. Data source: author's calculations using Library of Congress *Chronicling America*.

The estimates show that new factory mentions are followed by an increase in advertisements for skilled labor. Importantly, the increase is not limited to skilled manufacturing occupations. Skilled service and white-collar occupations also rise, consistent with the idea that manufacturing generates demand for organizational and administrative tasks. The help-wanted evidence therefore supports the occupational mechanism emphasized in the paper: manufacturing expansion is associated not only with greater demand for production-related skills, but also with greater demand for the skilled service and white-collar occupations that help organize production.

D Model Details, Equilibrium, and Wedge Implementation

This appendix collects technical material that is not included in the main text. Section D.1 defines the competitive equilibrium. Section D.2 derives the household fertility and education rules used in the quantitative model. Section D.3 presents a GDP per capita growth decomposition for the gross-output economy. Section **??** clarifies how the counterfactual economies with different manufacturing peaks are generated and explains the connection between changing the manufacturing demand weight w_m and introducing an explicit price or value-added-tax wedge.

D.1 Competitive equilibrium

A competitive equilibrium is a sequence of household choices,

$$\{C_{ij,t}, C_{i,t}, n_{i,t}, e_{i,t}\}_{i,j,t},$$

firm choices,

$$\{VA_{j,t}, Y_{j,t}, L_{s,j,t}, L_{u,j,t}, M_{a,j,t}, M_{m,j,t}, PS_{j,t}, HS_t, LS_t\}_{j,t},$$

producer-service task allocations,

$$\{x_{j,t}, z^*_{j,t}\}_{j,t},$$

prices,

$$\{p_{a,t}, p_{m,t}, p_{c,t}, w_{s,t}, w_{u,t}, p_{hs,t}, p_{ls,t}\}_t,$$

and aggregate states,

$$\{L_t, S_t, H_t, A_{j,t}\}_{j,t},$$

such that the following conditions hold.

First, households maximize utility subject to their budget and time constraints. Second, final-good producers choose value added, agricultural intermediates, manufacturing intermediates, and producer services to minimize costs. Third, producer-service tasks are assigned to the least-cost mode, implying the high-skill task shares characterized in Section 4. Fourth, all goods, labor, and intermediate-input markets clear. Fifth, aggregate states evolve according to the laws of motion for population, the skilled share, sectoral productivity, and codified organizational knowledge.

Manufacturing is the numeraire:

$$p_{m,t} = 1.$$

The skilled and unskilled labor-market clearing conditions are

$$S_t L_t = \sum_{j \in \{a,m,c\}} L_{s,j,t} + L_{s,hs,t},$$

and

$$(1 - S_t) L_t = \sum_{j \in \{a,m,c\}} L_{u,j,t} + L_{u,ls,t}.$$

Final-good market clearing requires gross output to equal final use plus intermediate use. For agriculture and manufacturing,

$$Y_{j,t} = C_{j,t} + \sum_{k \in \{a,m,c\}} M_{j,k,t}, \qquad j \in \{a, m\}.$$

Consumer services are used as final demand, while producer services enter production as intermediate inputs through the composite $PS_{j,t}$. Producer-service markets clear when aggregate demand for high- and low-skill producer-service modes equals the corresponding supplies HS_t and LS_t.

D.2 Household fertility and education rules

This subsection derives the closed-form fertility and education rules used in the quantitative implementation. For a household of origin $i \in \{a, m, c\}$, the fertility-education part of the problem can be written as

$$\max_{n_{i,t}, e_{i,t}} \quad (1 - \gamma) \log \left[1 - n_{i,t} \left(\tau_i + \tau_e(S_t) e_{i,t}\right)\right] + \gamma \log n_{i,t} + \gamma \log h_{i,t},$$

where

$$h_{i,t} = \frac{e_{i,t} + \tau_h}{e_{i,t} + \tau_h + g_t}.$$

Let

$$\chi_t \equiv \tau_e(S_t).$$

The first-order condition for fertility is

$$\frac{(1 - \gamma)(\tau_i + \chi_t e_{i,t})}{1 - n_{i,t}(\tau_i + \chi_t e_{i,t})} = \frac{\gamma}{n_{i,t}}.$$

Solving gives

$$n_{i,t} = \frac{\gamma}{\tau_i + \chi_t e_{i,t}}.$$

Substituting this expression into the household problem, the interior education choice solves

$$\frac{\partial \log h_{i,t}}{\partial e_{i,t}} = \frac{\chi_t}{\tau_i + \chi_t e_{i,t}}.$$

Using the human-capital function,

$$\frac{\partial \log h_{i,t}}{\partial e_{i,t}} = \frac{g_t}{(e_{i,t} + \tau_h)(e_{i,t} + \tau_h + g_t)}.$$

Thus the interior education choice satisfies

$$\frac{g_t}{(e_{i,t} + \tau_h)(e_{i,t} + \tau_h + g_t)} = \frac{\chi_t}{\tau_i + \chi_t e_{i,t}}.$$

Solving for $e_{i,t}$ yields

$$e^*_{i,t} = \sqrt{\frac{g_t(\tau_i - \chi_t \tau_h)}{\chi_t}} - \tau_h.$$

Imposing non-negativity,

$$e^*_{i,t} = \max\left\{ \sqrt{\frac{g_t(\tau_i - \chi_t \tau_h)}{\chi_t}} - \tau_h,\, 0 \right\}.$$

The two closed-form rules used in the model are therefore

$$n_{i,t} = \frac{\gamma}{\tau_i + \chi_t e^*_{i,t}},$$

and

$$e^*_{i,t} = \max\left\{ \sqrt{\frac{g_t(\tau_i - \chi_t \tau_h)}{\chi_t}} - \tau_h,\, 0 \right\}.$$

These expressions make explicit the demographic channel. Higher growth g_t raises the return to education, which increases education investment per child. Higher education investment raises the time cost of children and lowers fertility.

D.3 GDP per capita growth decomposition

This subsection derives a decomposition of GDP per capita growth in the gross-output economy. The decomposition is useful because the model differs from a standard value-added economy in two ways: sectors use intermediate inputs, and producer services can be reorganized across low- and high-skill modes.

Let real final output be

$$Y_t = F(C_{a,t}, C_{m,t}, C_{c,t}),$$

where $C_{j,t}$ is final use of good j. Let P_t be the final-output price index and $p_{j,t}$ the price of good j. Define final-demand expenditure shares as

$$\lambda_{j,t} \equiv \frac{p_{j,t} C_{j,t}}{P_t Y_t}.$$

Nominal final expenditure is $E_t = P_t Y_t$. Hence

$$\frac{Y_t}{L_t} = \frac{E_t/L_t}{P_t},$$

and

$$d\log\left(\frac{Y_t}{L_t}\right) = d\log\left(\frac{E_t}{L_t}\right) - d\log P_t.$$

Nominal income per person is

$$\frac{E_t}{L_t} = (1-\gamma)\left[S_t w_{s,t} + (1-S_t)w_{u,t}\right].$$

Let

$$\pi_t = \frac{w_{s,t}}{w_{u,t}}$$

denote the skill premium, and define

$$\Xi_t = 1 + S_t(\pi_t - 1).$$

Then

$$S_t w_{s,t} + (1-S_t)w_{u,t} = w_{u,t}\Xi_t,$$

so

$$d\log\left(\frac{E_t}{L_t}\right) = d\log w_{u,t} + \frac{\pi_t - 1}{\Xi_t} dS_t + \frac{S_t\pi_t}{\Xi_t} d\log\pi_t.$$

The term

$$\frac{\pi_t - 1}{\Xi_t} dS_t$$

is the direct contribution of skill accumulation to GDP per capita growth.

Now consider the price-index term. Gross output in sector $j \in \{a, m, c\}$ is

$$Y_{j,t} = A^G_{j,t} VA^{\theta_{j,t}}_{j,t} M^{\gamma_{a,j,t}}_{a,j,t} M^{\gamma_{m,j,t}}_{m,j,t} PS^{\gamma_{ps,j,t}}_{j,t},$$

where

$$\theta_{j,t} + \gamma_{a,j,t} + \gamma_{m,j,t} + \gamma_{ps,j,t} = 1.$$

The corresponding unit-cost equation is

$$p_{j,t} = \frac{1}{A^G_{j,t}} (c^V_{j,t})^{\theta_{j,t}} p^{\gamma_{a,j,t}}_{a,t} p^{\gamma_{m,j,t}}_{m,t} (c^{PS}_{j,t})^{\gamma_{ps,j,t}}.$$

Stack sectoral log prices in ℓ_t, gross-output productivities in a^G_t, value-added costs in v_t, and producer-service costs in q_t. Let G_t be the goods-input matrix, whose (i, j) entry is the cost share of good i used as an intermediate input by sector j. Define

$$\Theta_t = \operatorname{diag}(\theta_{a,t}, \theta_{m,t}, \theta_{c,t}), \qquad \Gamma^{PS}_t = \operatorname{diag}(\gamma_{ps,a,t}, \gamma_{ps,m,t}, \gamma_{ps,c,t}).$$

The stacked price system is

$$(I - G_t')\ell_t = -a_t^G + \Theta_t v_t + \Gamma_t^{PS} q_t.$$

Taking differentials gives

$$d\ell_t = (I - G_t')^{-1}\left[-da_t^G + \Theta_t dv_t + \Gamma_t^{PS} dq_t + d\Theta_t v_t + d\Gamma_t^{PS} q_t + dG_t'\ell_t\right].$$

Since

$$d\log P_t = \lambda_t' d\ell_t,$$

define the network-exposure vector

$$D_t' \equiv \lambda_t'(I - G_t')^{-1}.$$

Then

$$d\log P_t = D_t'\left[-da_t^G + \Theta_t dv_t + \Gamma_t^{PS} dq_t + d\Theta_t v_t + d\Gamma_t^{PS} q_t + dG_t'\ell_t\right].$$

The vector D_t is a Domar-style exposure vector. It measures the exposure of final demand to each gross-output sector, including both direct final use and indirect exposure through intermediate-input chains.

Let $g_t^G = d\log A_{j,t}^G$ denote common gross-output TFP growth. Let

$$g_t^A = d\log A_{a,t}^V, \qquad g_t^M = d\log A_{m,t}^V, \qquad g_t^S = d\log A_{c,t}^V$$

denote value-added productivity growth in agriculture, manufacturing, and services. Producer-service high- and low-skill mode productivities grow at the same service productivity growth rate g_t^S.

Define the service productivity exposure weight as

$$\Omega_t^S \equiv D_{c,t}\theta_{c,t} + \sum_{j\in\{a,m,c\}} D_{j,t}\gamma_{ps,j,t}.$$

The first term captures exposure to consumer-service value added. The second term captures exposure to producer services used as intermediate inputs by all final-good sectors.

Collect the terms associated with changing input-output shares into

$$\mathcal{R}_t \equiv D_t'\left[d\Theta_t v_t + d\Gamma_t^{PS} q_t + dG_t'\ell_t\right].$$

This term captures input-output reorganization: changes in value-added shares, producer-service input shares, and agriculture/manufacturing intermediate-input shares.

Collect the terms associated with changing producer-service task allocation into

$$\mathcal{X}_t \equiv \sum_{j\in\{a,m,c\}} D_{j,t}\gamma_{ps,j,t}\, d\log c_{j,t}^{PS}\big|_{\text{task reallocation}}.$$

This term captures the direct cost effect of moving producer-service tasks between low- and high-skill modes, holding mode prices fixed.

Combining the income and price-index terms, and using equilibrium labor-cost-share identities to cancel pure wage-price terms, gives

$$\boxed{\begin{aligned} d\log\left(\frac{Y_t}{L_t}\right) = &\underbrace{\left(\sum_j D_{j,t}\right) g_t^G + D_{a,t}\theta_{a,t}g_t^A + D_{m,t}\theta_{m,t}g_t^M + \Omega_t^S g_t^S}_{\text{network-weighted TFP growth}} \\ &+ \underbrace{\frac{\pi_t - 1}{\Xi_t} dS_t}_{\text{skill-composition growth}} - \underbrace{\mathcal{R}_t}_{\text{input-output reorganization}} - \underbrace{\mathcal{X}_t}_{\text{producer-service task reorganization}}. \end{aligned}}$$

This decomposition has three implications.

First, productivity growth is weighted by network exposure rather than by final expenditure or value-added shares alone. A sector matters for aggregate growth not only when it is consumed directly, but also when it is important as an intermediate supplier.

Second, the contribution of services has two components. Services enter directly through consumer-service value added, $D_{c,t}\theta_{c,t}g_t^S$, and indirectly through producer services used by all sectors,

$$\sum_j D_{j,t}\gamma_{ps,j,t}g_t^S.$$

Thus the relevant service-sector exposure is Ω_t^S, not only the final service expenditure share.

Third, structural change affects GDP per capita growth not only by changing sectoral employment shares, but also by changing production recipes and task allocations. If input-output reorganization lowers costs, then $\mathcal{R}_t < 0$ and the term $-\mathcal{R}_t$ raises growth. If producer-service task reorganization lowers costs, then $\mathcal{X}_t < 0$ and the term $-\mathcal{X}_t$ raises growth.

D.4 No-structural-change benchmark

The decomposition also clarifies the role of structural change. Consider a counterfactual that freezes final-demand shares, input-output coefficients, producer-service shares, and task allocations at their initial values:

$$\lambda_t = \lambda_0, \qquad G_t = G_0, \qquad \Theta_t = \Theta_0, \qquad \Gamma_t^{PS} = \Gamma_0^{PS}, \qquad x_{j,t} = x_{j,0}.$$

Then

$$D_0' = \lambda_0'(I - G_0')^{-1},$$

and

$$d\Theta_t = d\Gamma_t^{PS} = dG_t = 0, \qquad dx_{j,t} = 0.$$

Therefore

$$\mathcal{R}_t^{CF} = 0, \qquad \mathcal{X}_t^{CF} = 0.$$

The no-structural-change GDP per capita growth rate is

$$\begin{aligned} d\log\left(\frac{Y_t}{L_t}\right)^{CF} &= \left(\sum_j D_{j,0}\right) g_t^G + D_{a,0}\theta_{a,0} g_t^A + D_{m,0}\theta_{m,0} g_t^M \\ &\quad + \left[D_{c,0}\theta_{c,0} + \sum_j D_{j,0}\gamma_{ps,j,0}\right] g_t^S + \frac{\pi_t - 1}{\Xi_t} dS_t. \end{aligned}$$

The structural-change contribution is actual growth minus this counterfactual:

$$SC_t \equiv d\log\left(\frac{Y_t}{L_t}\right) - d\log\left(\frac{Y_t}{L_t}\right)^{CF}.$$

Using the decomposition above,

$$\begin{aligned} SC_t = &\left[\sum_j D_{j,t} - \sum_j D_{j,0}\right] g_t^G + \left[D_{a,t}\theta_{a,t} - D_{a,0}\theta_{a,0}\right] g_t^A \\ &+ \left[D_{m,t}\theta_{m,t} - D_{m,0}\theta_{m,0}\right] g_t^M \\ &+ \left[D_{c,t}\theta_{c,t} + \sum_j D_{j,t}\gamma_{ps,j,t} - D_{c,0}\theta_{c,0} - \sum_j D_{j,0}\gamma_{ps,j,0}\right] g_t^S \\ &- \mathcal{R}_t - \mathcal{X}_t. \end{aligned}$$

Structural change raises GDP per capita growth when it shifts network exposure toward high-growth productivity components or reorganizes production and tasks in a cost-saving way. It lowers growth when it shifts exposure toward low-growth components or raises the cost of production.

E Calibration Details and Counterfactual Implementation

This appendix provides additional details on the numerical calibration and clarifies the interpretation of the counterfactual variation in manufacturing peaks.

E.1 Numerical procedure

The model is simulated at an annual frequency and calibrated to the U.S. transition over 1850–2019. The simulation horizon is longer than the data window. The model date corresponding to 1850 is fixed at $t_0 = 60$, and all simulated moments are evaluated after applying this fixed calendar alignment. Gross-output input shares are interpolated so that the modern input-output structure is reached in calendar year 2025.

Let m^{data} denote the vector of empirical moments and $m^{model}(\Theta)$ the corresponding model moments. The calibrated parameter vector minimizes the loss

$$\mathcal{L}(\Theta) = \frac{1}{N} \sum_{\ell=1}^{N} \left(\frac{m_\ell^{model}(\Theta) - m_\ell^{data}}{\max\{|m_\ell^{data}|, \underline{s}_\ell\}} \right)^2,$$

for share and rate moments. Strictly positive level moments, such as normalized GDP per capita and the agriculture-to-service fertility ratio, are matched in logs. The constants $\underline{s}_\ell$ are moment-specific scale floors used to avoid overweighting moments close to zero.

The numerical minimization uses a global dual-annealing search followed by a Powell local search. I add small regularization penalties for high-frequency oscillations in the skill premium and in the aggregate skilled share. These penalties are not targeted economic moments; they rule out parameter vectors that fit isolated moments through implausibly volatile transition paths.

The half-saturation parameter in the codification frontier, $H_{1/2}$, is normalized relative to the simulated scale of knowledge accumulation. For each candidate parameter vector, I first run a probe simulation with a fixed input value of $H_{1/2}$. I then set $H_{1/2}$ equal to the average of the final observations of the probe H_t path, subject to lower and upper bounds. This prevents the arbitrary units of H_t from mechanically determining the location of the frontier.

E.2 Calibrated parameters and targeted moments

The internally calibrated parameter vector is

$$\Theta = \left(\gamma, \tau_a, \phi_a, \phi_m, \phi_c, g^{GO}, \kappa_a, \kappa_m, \kappa_c, \eta_H, c_{col}, \alpha_a, \alpha_m, \alpha_c, \varepsilon_a, \varepsilon_c, w_a, w_c\right).$$

The manufacturing demand weight is fixed at $w_m = 3.5$. Manufacturing is the numeraire and $\varepsilon_m = 1$.

The targeted moments are: sectoral employment shares in agriculture, manufacturing, and services; the peak manufacturing employment share; the fertility path; the agriculture-to-service fertility ratio; normalized GDP per capita; long-run value-added productivity growth in agriculture, manufacturing, and services; the aggregate skilled employment share; within-value-added skilled employment shares by sector; high-skill employment shares in producer-service inputs by using sector; and the aggregate change in the high-skill producer-service share between 1960 and 2019. The calibration is overidentified: the number of targeted moments exceeds the number of internally calibrated parameters.

E.3 High-skill producer-service inputs by final sector

This subsection describes the construction of the high-skill producer-service moments used in the calibration. These moments discipline the sector-specific codification parameters κ_j, which determine how easily producer-service tasks used by final sector $j \in \{a, m, c\}$ can be reorganized into high-skill form.

The object I need is not the average education of all workers in agriculture, manufacturing, or consumer services. The model distinguishes final-good production from the producer-service tasks used as intermediate inputs. The relevant empirical counterpart is therefore the education intensity of producer-service occupations employed within each broad final sector. In the model, these occupations correspond to managerial, professional, technical, administrative, accounting, engineering, planning, and coordination tasks.

I construct the U.S. moments using the 2019 IPUMS USA Census sample. I classify workers into broad industries using the 1950 Census industry classification. Agriculture includes agriculture, forestry, and fisheries. Manufacturing includes durable and nondurable manufacturing. Consumer services include retail, personal services, entertainment, education, health, welfare, religious services, and other consumer-facing service industries. I then identify producer-service occupations using the 1950 Census occupation classification. The baseline definition includes occupations that clearly perform professional, technical, managerial, or organizational tasks, such as accountants, architects, engineers, lawyers, statisticians, personnel workers, technicians, purchasing agents, inspectors, insurance agents, real-estate agents, and related business-facing occupations. The replication files report the exact industry and occupation codes.

For each broad final sector j, I compute the weighted share of producer-service workers with at least one year of college:

$$HSPS_j = \frac{\sum_{n \in j} \omega_n \mathbb{1}\{\text{PS occupation}_n = 1\} \mathbb{1}\{\text{college}_n = 1\}}{\sum_{n \in j} \omega_n \mathbb{1}\{\text{PS occupation}_n = 1\}},$$

where ω_n is the person weight. These are the empirical counterparts of the high-skill shares within producer-service inputs used by agriculture, manufacturing, and consumer services in the model. The resulting U.S. moments are

$$HSPS_a = 0.393, \qquad HSPS_m = 0.653, \qquad HSPS_c = 0.529.$$

Thus, producer-service occupations employed in manufacturing are substantially more college-intensive than producer-service occupations employed in agriculture or consumer services. This ranking is the key moment used to discipline the ordering of codification resistance in the model. The calibrated model chooses $\kappa_m < \kappa_c < \kappa_a$, so manufacturing-related producer-service tasks are easiest to reorganize into high-skill form.

I also examine whether this pattern is specific to the United States. Using IPUMS International, I construct the same object for 298 census samples spanning countries at very different stages of development, including low, middle and high income countries. In then repeat the exercise by classifying industries into agriculture, manufacturing, and consumer services using the harmonized IPUMS industry variable. I classify producer-service occupations using ISCO major groups corresponding to managerial, professional, and technical occupations.[8] I then compute college shares separately by broad sector and occupation type. Figure 23 reports the resulting averages for agriculture, manufacturing, and consumer services. Producer-service occupations are much more college-intensive than other occupations in every broad sector. Among producer-service occupations, the college share is also higher in manufacturing than in consumer services or agriculture. This supports the interpretation of the U.S. calibration moments: the relevant margin is not that all manufacturing workers are uniformly more educated, but that manufacturing uses a set of producer-service occupations that is especially skill-intensive.

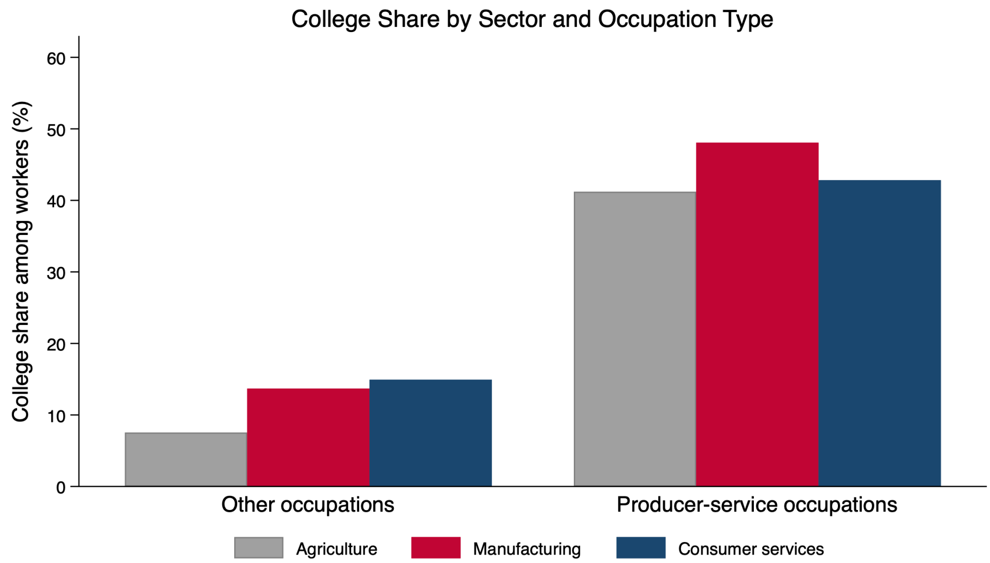

Figure 23: College share by sector and occupation type in IPUMS International. The figure uses 298 census samples covering economies across the development spectrum. Producer-service occupations are defined using managerial, professional, and technical occupation groups. Bars report average college shares separately for other occupations and producer-service occupations within agriculture, manufacturing, and consumer services.

8. This corresponds to ISCO code 3 or less.

E.4 Untargeted producer-service response

As an external validation of the producer-service block, I compare the model to a final-demand-shock exercise in the 2015 OECD ICIO table. In the data, I compute the service output induced by a \$100 final-demand shock to agriculture, manufacturing, and consumer services. In the model, I apply the analogous shock to the calibrated economy and compute the high-skill share of induced producer-service output. This moment is not directly targeted in the calibration.

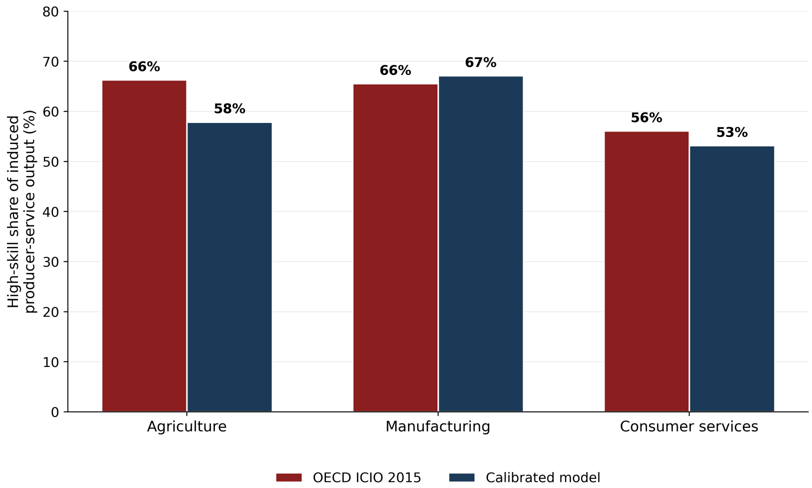

Figure 24: High-skill share of producer-service output induced by a \$100 final-demand shock. The data exercise uses Leontief total requirements from the 2015 OECD ICIO table; the model exercise applies the same shock to the calibrated economy.

E.5 Interpreting the manufacturing-demand counterfactual

The counterfactuals in the main text vary the manufacturing demand shifter w_m to generate economies with different peak manufacturing employment shares. This is a reduced-form way to change the relative pull of manufacturing during the transition while holding technology, fertility and education preferences, and production primitives fixed.

This counterfactual has an equivalent price-wedge representation. Final demand over agriculture, manufacturing, and consumer services implies relative demand terms of the form

$$K_a = \left(\frac{p_a}{p_m}\right)^{1-\sigma} Y^{(1-\sigma)(k_a-1)} \frac{w_a}{w_m^{k_a}}, \qquad K_c = \left(\frac{p_c}{p_m}\right)^{1-\sigma} Y^{(1-\sigma)(k_c-1)} \frac{w_c}{w_m^{k_c}},$$

where $k_j \equiv \varepsilon_j/\varepsilon_m$. Changing the manufacturing demand weight from w_m to $f_m w_m$ therefore changes the relative demand terms to

$$K_a(f_m) = K_a(1) f_m^{-k_a}, \qquad K_c(f_m) = K_c(1) f_m^{-k_c}.$$

Now consider an economy with the original w_m, but in which households face effective final-good prices

$$\widetilde{p}_{a,t} = (1+\tau_a)p_{a,t}, \qquad \widetilde{p}_{m,t} = p_{m,t}, \qquad \widetilde{p}_{c,t} = (1+\tau_c)p_{c,t}.$$

The wedge economy generates the same relative demand schedules as the $f_m w_m$ economy if

$$(1+\tau_a)^{1-\sigma} = f_m^{-k_a}, \qquad (1+\tau_c)^{1-\sigma} = f_m^{-k_c}.$$

Equivalently,

$$\boxed{1+\tau_a = f_m^{-k_a/(1-\sigma)}, \qquad 1+\tau_c = f_m^{-k_c/(1-\sigma)}.}$$

Thus the w_m counterfactual can be interpreted either as a manufacturing demand shifter or as an equivalent system of final-good wedges. The two representations are algebraically identical for household final-demand allocations. The purpose of varying w_m is therefore not to assign structural content to tastes, but to provide a parsimonious parameterization of a relative-price distortion that changes the size of the manufacturing phase.

E.6 Construction of input-output coefficients

This subsection describes how I construct the modern input-output coefficients used in the calibration. The model production function for each final-good sector $j \in \{a, m, c\}$ uses value added, agricultural intermediates, manufacturing intermediates, and producer-service intermediates:

$$Y_{j,t} = A_{go,j,t} VA_{j,t}^{\theta_{j,t}} M_{a,j,t}^{\gamma_{a,j,t}} M_{m,j,t}^{\gamma_{m,j,t}} PS_{j,t}^{\gamma_{ps,j,t}}, \qquad \theta_{j,t} + \gamma_{a,j,t} + \gamma_{m,j,t} + \gamma_{ps,j,t} = 1.$$

The modern - 2025 - coefficients are constructed from the BEA Use Table, after redefinitions, at purchasers' prices. I use the sector-level table and map BEA industries into the model sectors as follows. Agriculture is BEA sector 11, manufacturing is BEA sector $31G$, and consumer services are the sum of educational services, health care, and social assistance; arts, entertainment, recreation, accommodation, and food services; and other services except government, corresponding to BEA sectors 6, 7, and 81. Producer-service inputs are the sum of transportation and warehousing, information, finance, insurance, real estate, rental, and leasing, and professional and business services, corresponding to BEA sectors $48TW$, 51, $FIRE$, and $PROF$.

For each destination sector j, I compute gross output Y_j and four raw shares:

$$\theta_j^{raw} = \frac{VA_j}{Y_j}, \qquad \gamma_{a,j}^{raw} = \frac{U_{A,j}}{Y_j}, \qquad \gamma_{m,j}^{raw} = \frac{U_{M,j}}{Y_j}, \qquad \gamma_{ps,j}^{raw} = \frac{U_{PS,j}}{Y_j},$$

where $U_{A,j}$ is the use of agricultural inputs by destination sector j, $U_{M,j}$ is the use of manufacturing inputs, and $U_{PS,j}$ is the use of producer-service inputs. Since the model abstracts from residual input categories such as mining, utilities, construction, government, scrap, and noncomparable imports, I normalize the four model-relevant shares to sum to one:

$$(\theta_j, \gamma_{a,j}, \gamma_{m,j}, \gamma_{ps,j}) = \frac{\left(\theta_j^{raw}, \gamma_{a,j}^{raw}, \gamma_{m,j}^{raw}, \gamma_{ps,j}^{raw}\right)}{\theta_j^{raw} + \gamma_{a,j}^{raw} + \gamma_{m,j}^{raw} + \gamma_{ps,j}^{raw}}.$$

For agriculture, the BEA table gives

$$\theta_a^{raw} = 0.394, \qquad \gamma_{a,a}^{raw} = 0.295, \qquad \gamma_{m,a}^{raw} = 0.222, \qquad \gamma_{ps,a}^{raw} = 0.051.$$

After normalizing the four model-relevant components, the coefficients are

$$\theta_a = 0.410, \qquad \gamma_{a,a} = 0.307, \qquad \gamma_{m,a} = 0.230, \qquad \gamma_{ps,a} = 0.053.$$

For manufacturing, the BEA table gives

$$\theta_m^{raw} = 0.354, \qquad \gamma_{a,m}^{raw} = 0.060, \qquad \gamma_{m,m}^{raw} = 0.407, \qquad \gamma_{ps,m}^{raw} = 0.070.$$

After normalizing the four model-relevant components, the unadjusted coefficients are

$$\theta_m^{BEA} = 0.397, \qquad \gamma_{a,m}^{BEA} = 0.068, \qquad \gamma_{m,m}^{BEA} = 0.457, \qquad \gamma_{ps,m}^{BEA} = 0.078.$$

This raw producer-service share captures producer services purchased from outside the manufacturing sector, but it does not fully capture service tasks performed inside manufacturing establishments.

I therefore make a conservative adjustment for in-house producer-service activity in manufacturing. A growing literature emphasizes that the boundary between manufacturing and services is blurred because manufacturing firms both purchase services from outside suppliers and perform many service tasks internally (Miroudot and Cadestin, 2017; Bernard et al., 2017). Miroudot and Cadestin (2017) report that, across countries, between 25 and 60 percent of employment in manufacturing firms is in service-support functions such as R&D, engineering, transport, logistics, distribution, marketing, sales, after-sale services, IT, management, and back-office support. To capture this margin without mechanically assigning a large share of manufacturing to producer services, I reallocate only 20 percent of the normalized manufacturing value-added share and 20 percent of the normalized manufacturing-intermediate share to producer services:

$$\theta_m = 0.8\,\theta_m^{BEA}, \qquad \gamma_{m,m} = 0.8\,\gamma_{m,m}^{BEA},$$
$$\gamma_{ps,m} = \gamma_{ps,m}^{BEA} + 0.2\,\theta_m^{BEA} + 0.2\,\gamma_{m,m}^{BEA}, \qquad \gamma_{a,m} = \gamma_{a,m}^{BEA}.$$

This gives the adjusted manufacturing coefficients

$$\theta_m = 0.318, \qquad \gamma_{a,m} = 0.068, \qquad \gamma_{m,m} = 0.365, \qquad \gamma_{ps,m} = 0.249.$$

The adjustment is conservative relative to the OECD evidence: the resulting producer-service share in manufacturing is about 25 percent, which is at the lower end of the 25–60 percent range documented for service-support employment inside manufacturing firms.

For consumer services, the BEA table gives

$$\theta_c^{raw} = 0.613, \qquad \gamma_{a,c}^{raw} = 0.002, \qquad \gamma_{m,c}^{raw} = 0.108, \qquad \gamma_{ps,c}^{raw} = 0.224.$$

After normalizing the four model-relevant components, the coefficients are

$$\theta_c = 0.647, \qquad \gamma_{a,c} = 0.002, \qquad \gamma_{m,c} = 0.114, \qquad \gamma_{ps,c} = 0.237.$$

The final modern input-output coefficients used in the calibration are therefore

Destination sector	θ_j	$\gamma_{a,j}$	$\gamma_{m,j}$	$\gamma_{ps,j}$
Agriculture	0.410	0.307	0.230	0.053
Manufacturing	0.318	0.068	0.365	0.249
Consumer services	0.647	0.002	0.114	0.237

In the calibrated transition, the economy starts in 1850 with value-added production, $\theta_{j,1850} = 1$, and the input-output coefficients gradually converge to these modern values by 2025. This allows the model to capture the increasing role of intermediate inputs and producer services over development while preserving a parsimonious three-sector structure.

F Policy Extensions

This appendix studies how trade policy affects the development path in the calibrated economy. The main counterfactuals in Section 6 vary the size of the manufacturing phase directly. Here I instead introduce an explicit trade environment and ask whether openness amplifies or weakens the manufacturing-based development mechanism.

F.1 Trade environment

I extend the calibrated economy to a small-open-economy setting. Home can trade agriculture and manufacturing with a rest-of-world economy. Consumer services and producer services remain non-traded. The rest of the world follows the same calibrated transition as Home, but may be shifted forward or backward in time. Let Δ denote Home's development head start relative to the rest of the world. If $\Delta > 0$, Home is more advanced than its trading partner; if $\Delta < 0$, Home trades with a more advanced partner.

Trade is introduced through an Armington aggregator for the two traded final goods. For $j \in \{a, m\}$, the consumer price of the traded composite is

$$p_{j,t}^{A} = \left[(1-\lambda_j) p_{j,t}^{1-\theta} + \lambda_j (\tau p_{j,t}^{*})^{1-\theta} \right]^{\frac{1}{1-\theta}},$$

where $p_{j,t}$ is the Home producer price, $p_{j,t}^{*}$ is the foreign producer price, τ is the iceberg trade cost, θ is the Armington elasticity, and λ_j governs the weight on foreign varieties. The domestic expenditure share is

$$d_{j,t} = (1-\lambda_j) \left(\frac{p_{j,t}}{p_{j,t}^{A}} \right)^{1-\theta}.$$

Before the opening date, trade costs are prohibitive and $d_{j,t} = 1$. After the opening date, trade costs fall to the counterfactual level.

The production side is kept close to the closed-economy model. Domestic production responds to domestic absorption, not to foreign demand. Thus, trade affects employment and dynamics by changing the domestic expenditure share on Home agriculture and manufacturing. Exports are computed as an accounting residual from domestic gross output net of domestic final absorption. Imports are then scaled so that the value of imports equals the value of exports. Hence trade is balanced period by period:

$$\sum_{j \in \{a,m\}} M_{j,t} = \sum_{j \in \{a,m\}} X_{j,t}.$$

This balanced-trade restriction prevents the welfare effects from being driven by external borrowing or lending. It also makes the exercise conservative: trade affects the development path only through prices, domestic absorption, and sectoral reallocation.

F.2 Sectoral effects of opening

Figure 25 shows the first-order effect of the benchmark trade opening. Relative to the protected counterfactual, openness reduces agriculture and manufacturing employment and increases service employment. This is the key margin in the model. Trade does not matter only because it changes prices; it changes the sectoral path through which the economy develops.

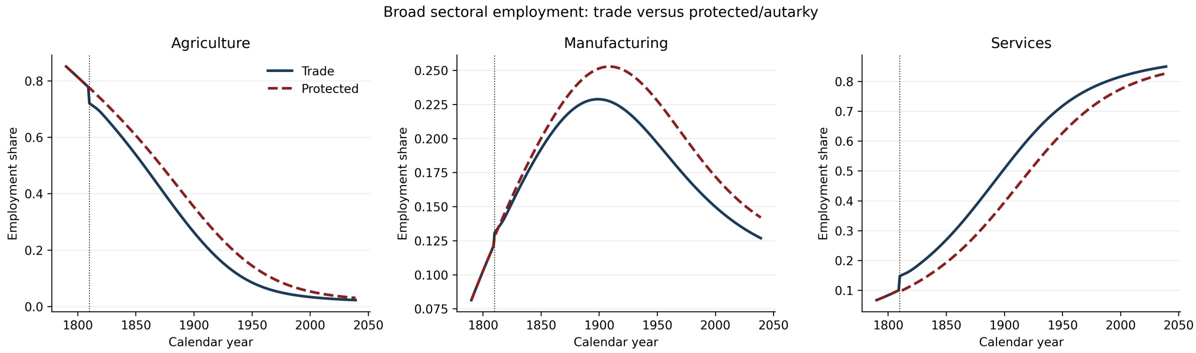

Figure 25: Broad sectoral employment under trade and protection.

Figure 26 shows that the service expansion induced by trade is also compositionally different. In the open economy, services expand more toward consumer services and less toward producer services. Within producer services, the economy also develops a lower high-skill producer-service share for much of the transition. This is the relevant dynamic channel: trade moves labor into services, but into a less skill-intensive service economy.

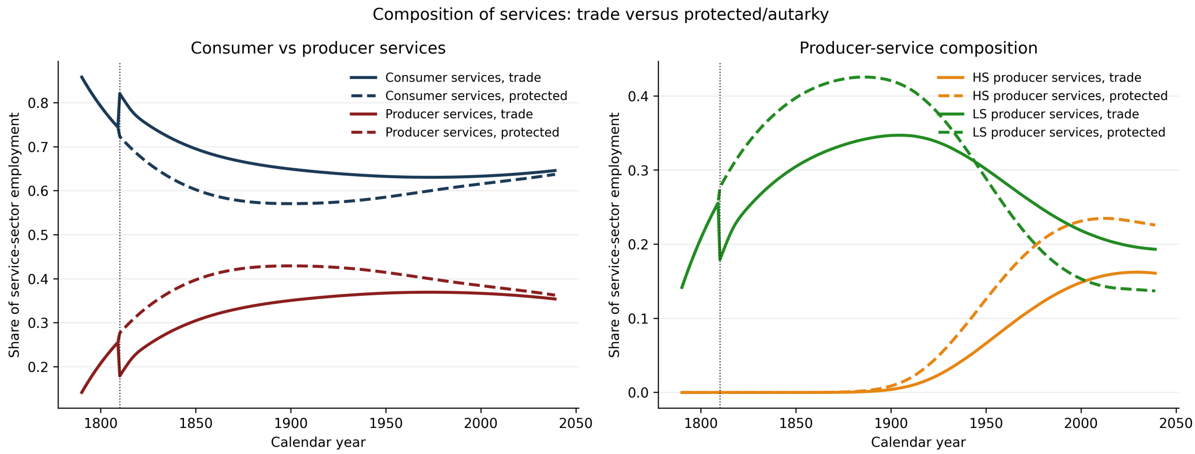

Figure 26: Composition of services under trade and protection.

This prediction is directly connected to the empirical validation exercise discussed later in the paper. Using subnational trade shocks constructed from IPUMS International, I test whether trade-induced reductions in manufacturing exposure shift local labor markets toward less producer-service-intensive and less skill-intensive service employment.

F.3 Welfare gains from trade

I compute welfare gains as the difference between the open economy and the protected counterfactual:

$$\Delta W_t = W_t^{open} - W_t^{protected}.$$

The total discounted gain is summarized as a consumption-equivalent variation:

$$\text{CEV} = \exp\left(\frac{\sum_t \beta^t \Delta W_t}{(1-\gamma)\sum_t \beta^t}\right) - 1.$$

This expresses the average discounted welfare gain as a permanent proportional increase in consumption.

Figure 27 plots the dynamic welfare path in the benchmark opening. Trade raises welfare on impact because households gain access to cheaper traded goods. Over time, however, the welfare gain declines and eventually turns negative. The reason is that the static consumption gain is partly offset by a weaker manufacturing phase and a less skill-intensive service transition.

Figure 27: Dynamic gains from trade in the benchmark opening.

Thus, the model separates two forces. The static gains from trade are positive at the opening date. The dynamic effect can be negative if trade reduces the manufacturing phase that builds human capital, codified organizational knowledge, and high-skill producer-service capacity.

F.4 Manufacturing comparative advantage

I next vary Home's static comparative advantage in manufacturing. The exercise changes the gross-output productivity wedge

$$\frac{A_m^{GO}}{A_a^{GO}},$$

holding fixed the dynamic spillover parameters. Hence the exercise changes Home's static specialization pattern without mechanically changing the strength of manufacturing spillovers.

Figure 28 shows that the gains from trade rise sharply with Home's static manufacturing comparative advantage. When Home has weak manufacturing comparative advantage, the gains from trade are small or negative. When Home has strong manufacturing comparative advantage, the CEV becomes large. Figure 29 confirms that the wedge works through the intended channel: stronger manufacturing comparative advantage raises export intensity in tradables.

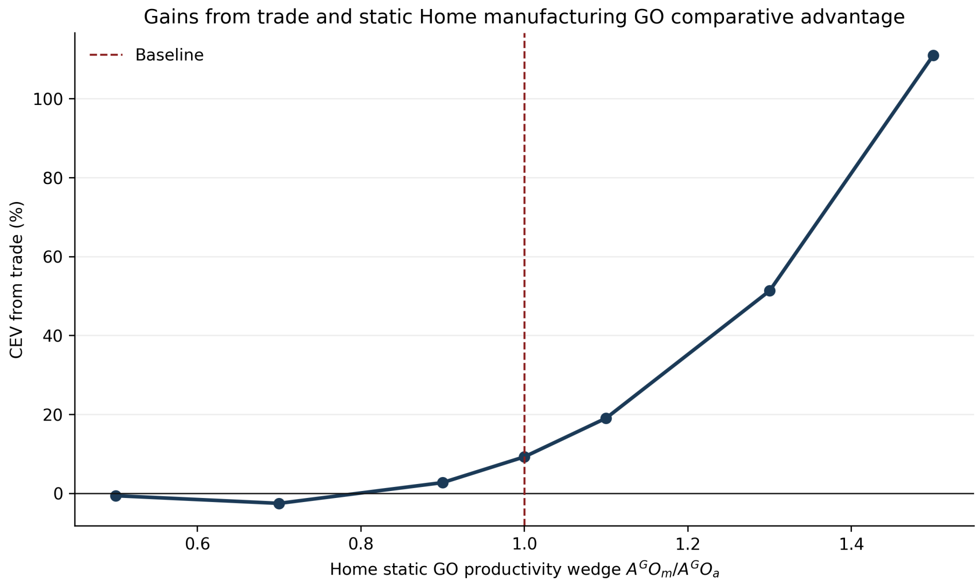

Figure 28: Gains from trade and static Home manufacturing comparative advantage.

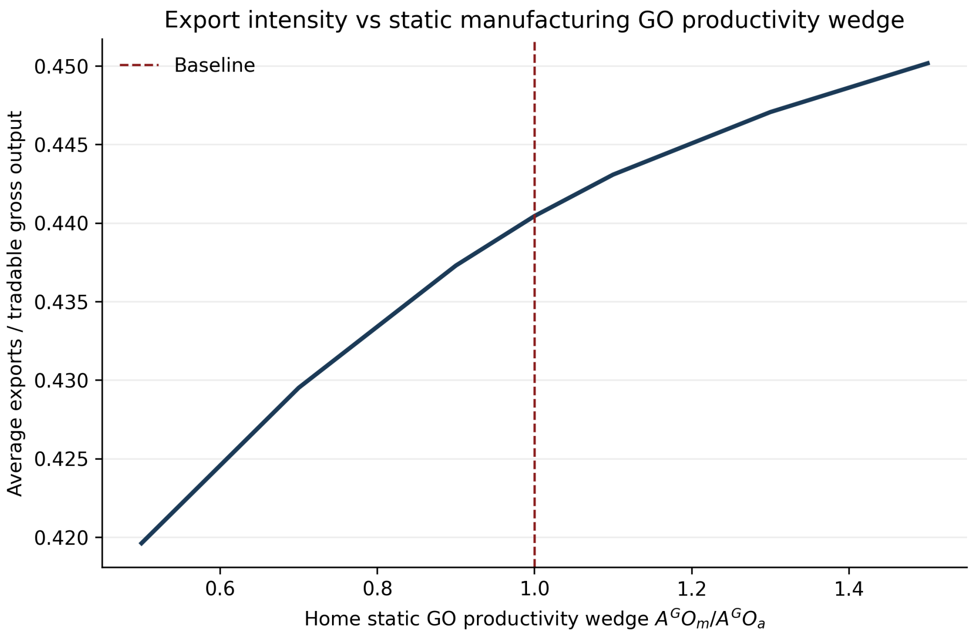

Figure 29: Export intensity and static manufacturing comparative advantage.

The lesson is that openness is most beneficial when it supports, rather than displaces, the manufacturing phase. Comparative advantage in manufacturing makes trade less likely to generate premature movement into low-skill services.

F.5 Development gaps and timing

Finally, I vary both the timing of trade opening and the development gap between Home and the rest of the world. Figure 30 reports two heatmaps. The left panel reports total discounted gains from trade. The right panel reports the long-run dynamic gain, measured by the final value of $W_t^{open} - W_t^{protected}$.

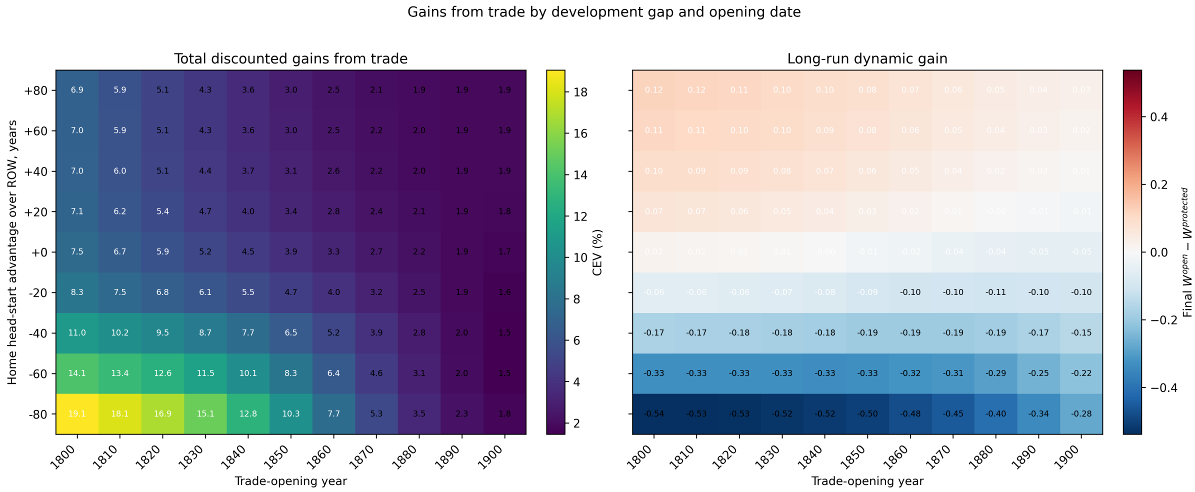

Figure 30: Gains from trade by development gap and opening date. Positive values on the vertical axis mean that Home is more advanced than the rest of the world. Negative values mean that Home opens to a more advanced partner.

The two panels reveal a sharp distinction between total and dynamic gains. Total discounted gains are largest when Home opens early to a more advanced partner. This reflects the large static benefits from cheaper traded goods. Long-run dynamic gains display the opposite pattern. They are positive when Home is more advanced than its partner, and negative when Home opens to a more advanced partner.

This asymmetry follows directly from the model mechanism. A less advanced economy opening to a more advanced partner imports cheaper manufactures, contracts domestic manufacturing, and enters services earlier. This raises consumption in the short run but weakens the domestic accumulation of high-skill producer-service capabilities. By contrast, an advanced economy opening to a less advanced partner preserves its manufacturing and producer-service base while still benefiting from trade. In that case, the long-run dynamic welfare effect can remain positive.

These results should however not be read as an argument for autarky. This exercise abstracts from technology diffusion, foreign direct investment, imported capital goods, and learning through exporting. Instead, we isolate one force: when manufacturing builds the human-capital and organizational foundations of high-growth services, trade policy affects development not only through prices, but also through the type of service economy that emerges.